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Lai et al.

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(45) Date of Patent: Aug. 12, 2025

(54) **DIPOLE-ENGINEERED HIGH-K GATE DIELECTRIC AND METHOD FORMING SAME**

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(73) Assignee: **Taiwan Semiconductor Manufacturing Co., Ltd.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/665,199**

(22) Filed: **May 15, 2024**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Continuation of application No. 18/356,860, filed on Jul. 21, 2023, now Pat. No. 12,020,941, which is a

(Continued)

(51) **Int. Cl.**

H01L 21/28 (2025.01)
H01L 21/3115 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***H01L 21/28185*** (2013.01); ***H01L 21/3115*** (2013.01); ***H10D 64/685*** (2025.01);

(Continued)

(58) **Field of Classification Search**

CPC H01L 21/31111; H01L 27/092; H01L 21/823462; H01L 29/66795;

(Continued)

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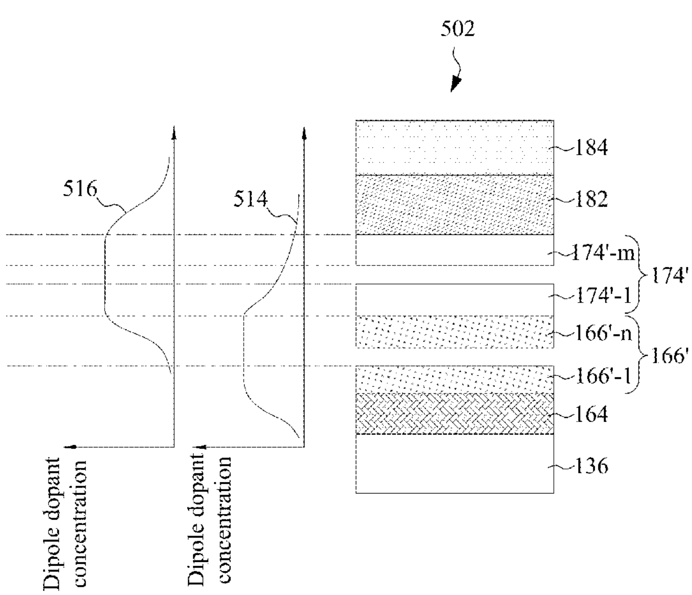
Primary Examiner — Mouloucoulaye Inoussa
(74) *Attorney, Agent, or Firm* — Slater Matsil, LLP

(57) **ABSTRACT**

A method includes forming an oxide layer on a semiconductor region, and depositing a first high-k dielectric layer over the oxide layer. The first high-k dielectric layer is formed of a first high-k dielectric material. The method further includes depositing a second high-k dielectric layer over the first high-k dielectric layer, wherein the second high-k dielectric layer is formed of a second high-k dielectric material different from the first high-k dielectric material, depositing a dipole film over and contacting a layer selected from the first high-k dielectric layer and the second high-k dielectric layer, performing an annealing process to drive-in a dipole dopant in the dipole film into the layer, removing the dipole film, and forming a gate electrode over the second high-k dielectric layer.

20 Claims, 28 Drawing Sheets



Related U.S. Application Data

division of application No. 17/094,241, filed on Nov. 10, 2020, now Pat. No. 11,784,052.

(60) Provisional application No. 63/031,099, filed on May 28, 2020.

(51) **Int. Cl.**

H10D 64/68	(2025.01)
H10D 84/01	(2025.01)
H10D 84/03	(2025.01)
H10D 84/83	(2025.01)

(52) **U.S. Cl.**
CPC ***H10D 84/0144*** (2025.01); ***H10D 84/038*** (2025.01); ***H10D 84/834*** (2025.01); *H10D 84/0158* (2025.01)

(58) **Field of Classification Search**
CPC H01L 27/0886; H01L 29/513; H01L 21/3115; H01L 21/28185; H01L 21/823857; H01L 27/088; H01L 21/823431; H01L 21/823821; H01L 29/78; H01L 21/28088; H01L 29/66545; H01L 29/401; H01L 29/518; H01L 29/517; H01L 21/02183; H01L 21/02321; H01L 21/0228; H01L 29/4966; H01L 21/28176; H01L 29/0657; H01L 21/283; H01L 21/324; H01L 29/42356; H01L 29/785; H01L 21/32051; H01L 21/3215; H01L 21/28097; H01L 21/32055; H01L 29/51; H01L 29/4975; H01L 21/823475; H01L 29/0847; H01L 21/76802; H01L 21/823864; H01L 21/823871; H01L 21/823418; H01L 27/0203; H01L 29/665; H01L 21/30604; H01L 21/823814; H01L 27/1104; H01L 21/26513; H01L 27/0922; H01L 27/0924; H01L 21/76814; H01L 21/76897; H01L 27/0928; H01L 27/1116; H01L 23/485; H01L 21/76895; H01L 29/41791; H01L 21/76816; H01L 21/76846; H01L 23/53252; H01L 23/53209; H01L 23/53238; H01L 23/53266; H01L 23/53295; H01L 21/823807; H01L 29/42376; H01L 21/823456; H01L 29/4958; H01L 21/28247; H01L 21/28008; H01L 21/823842; H01J 37/32

See application file for complete search history.

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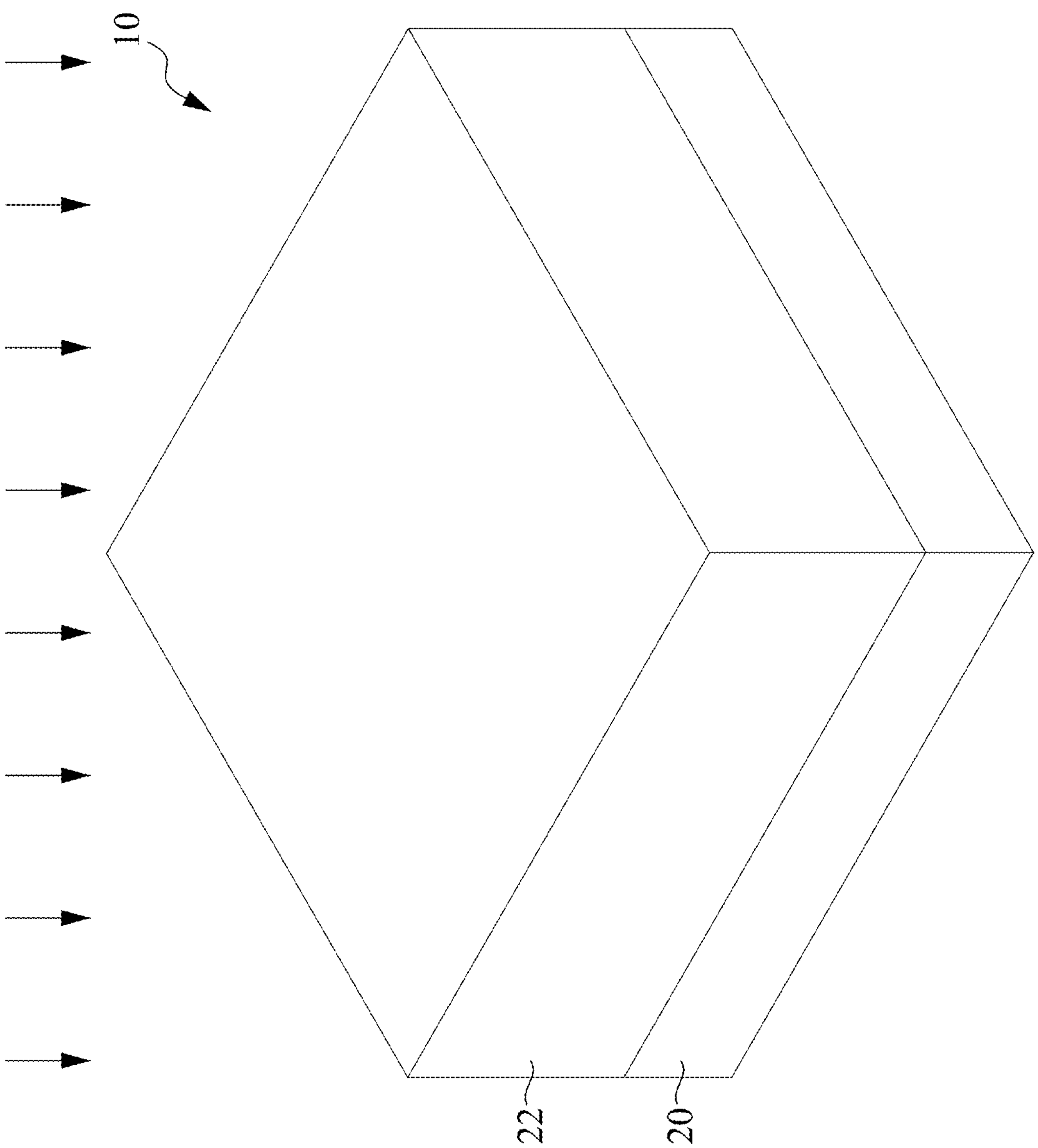


FIG. 1

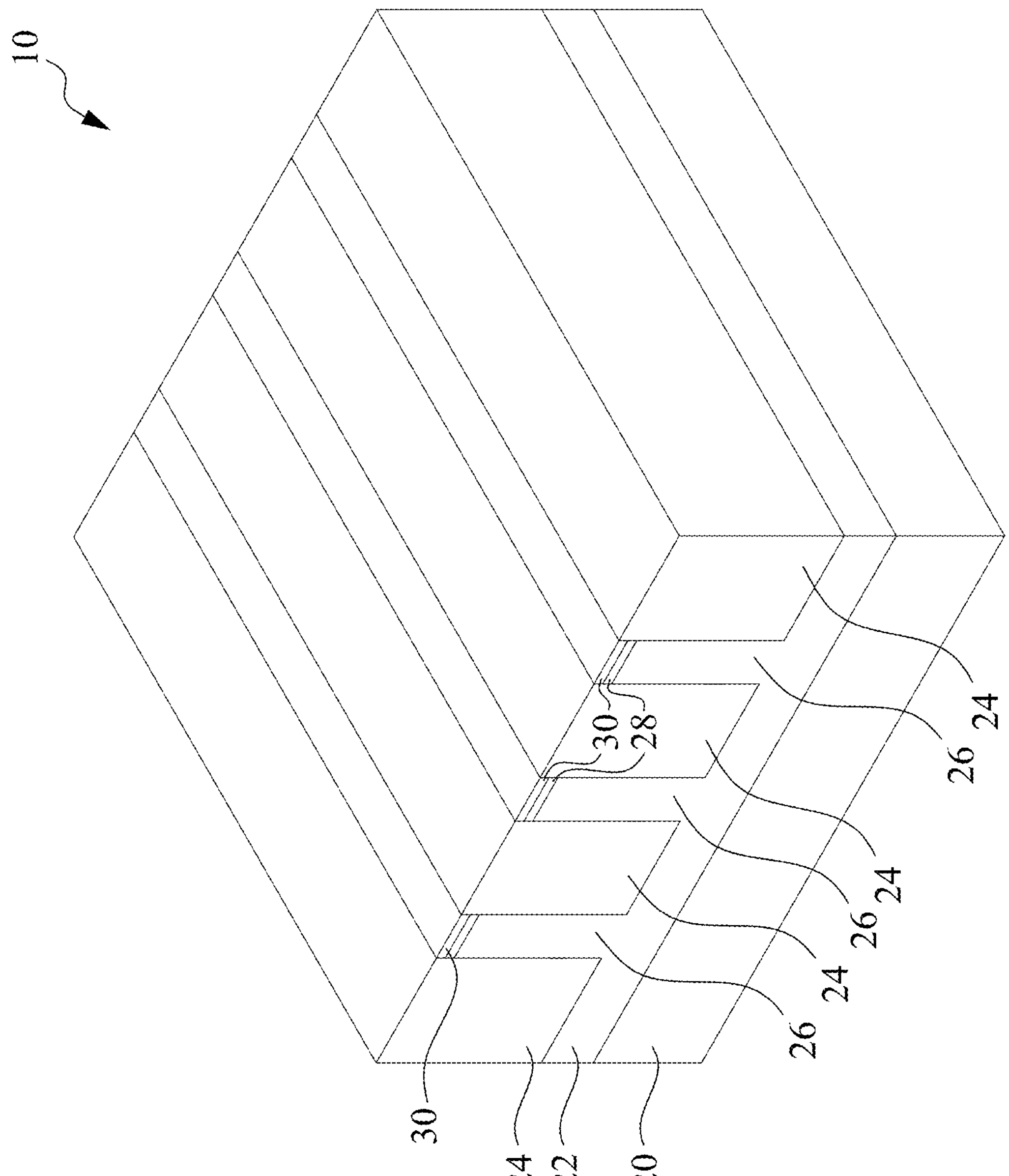


FIG. 2

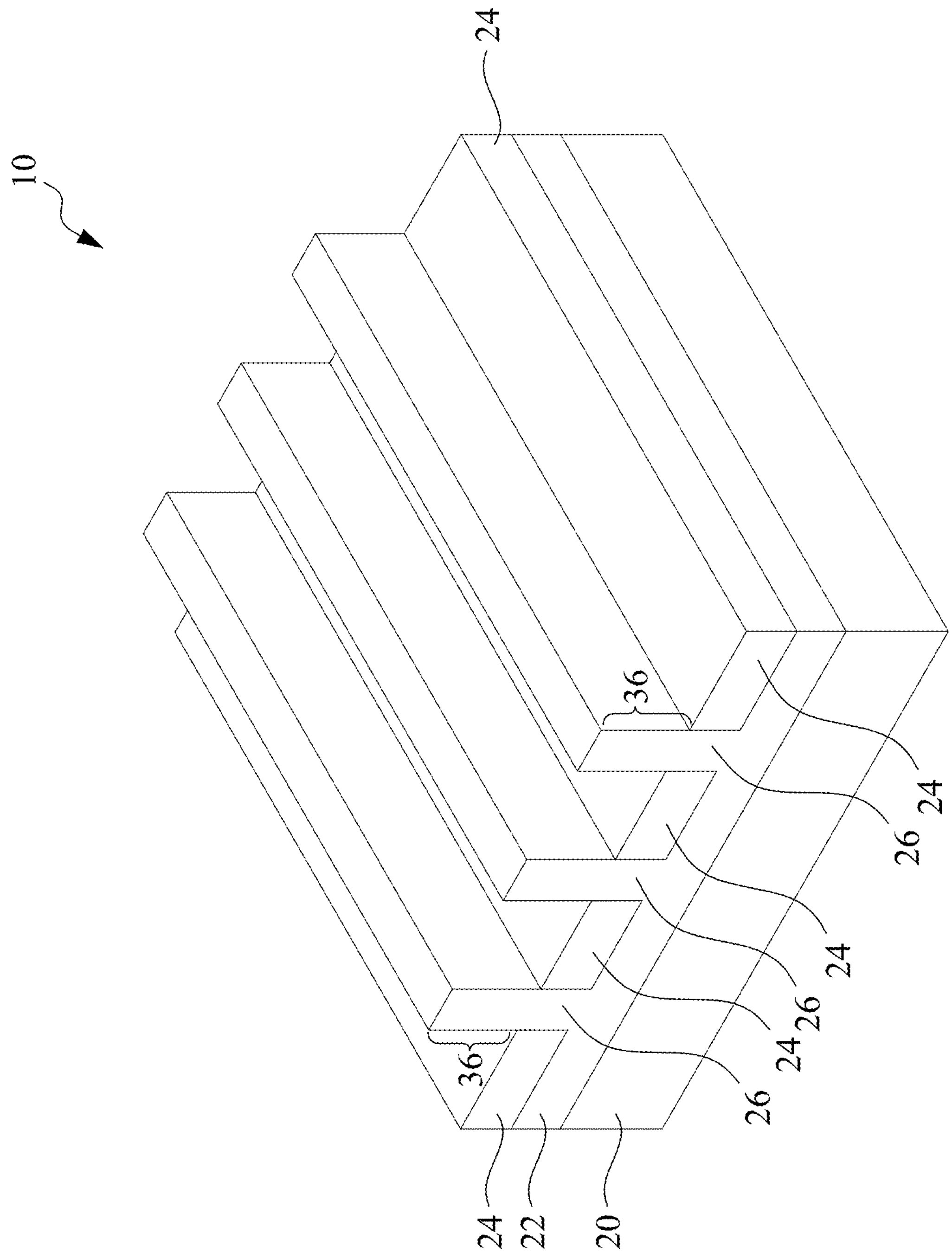
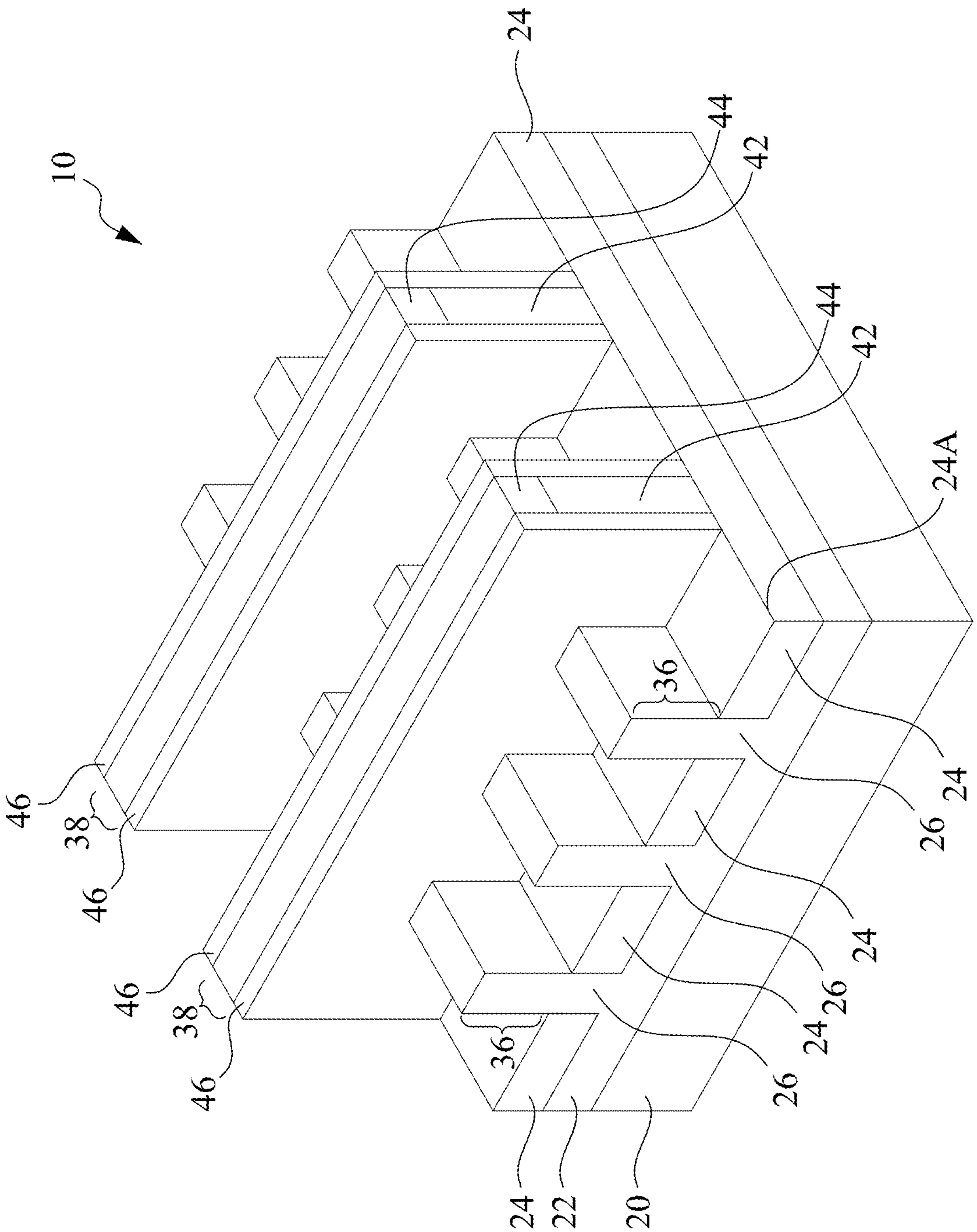


FIG. 3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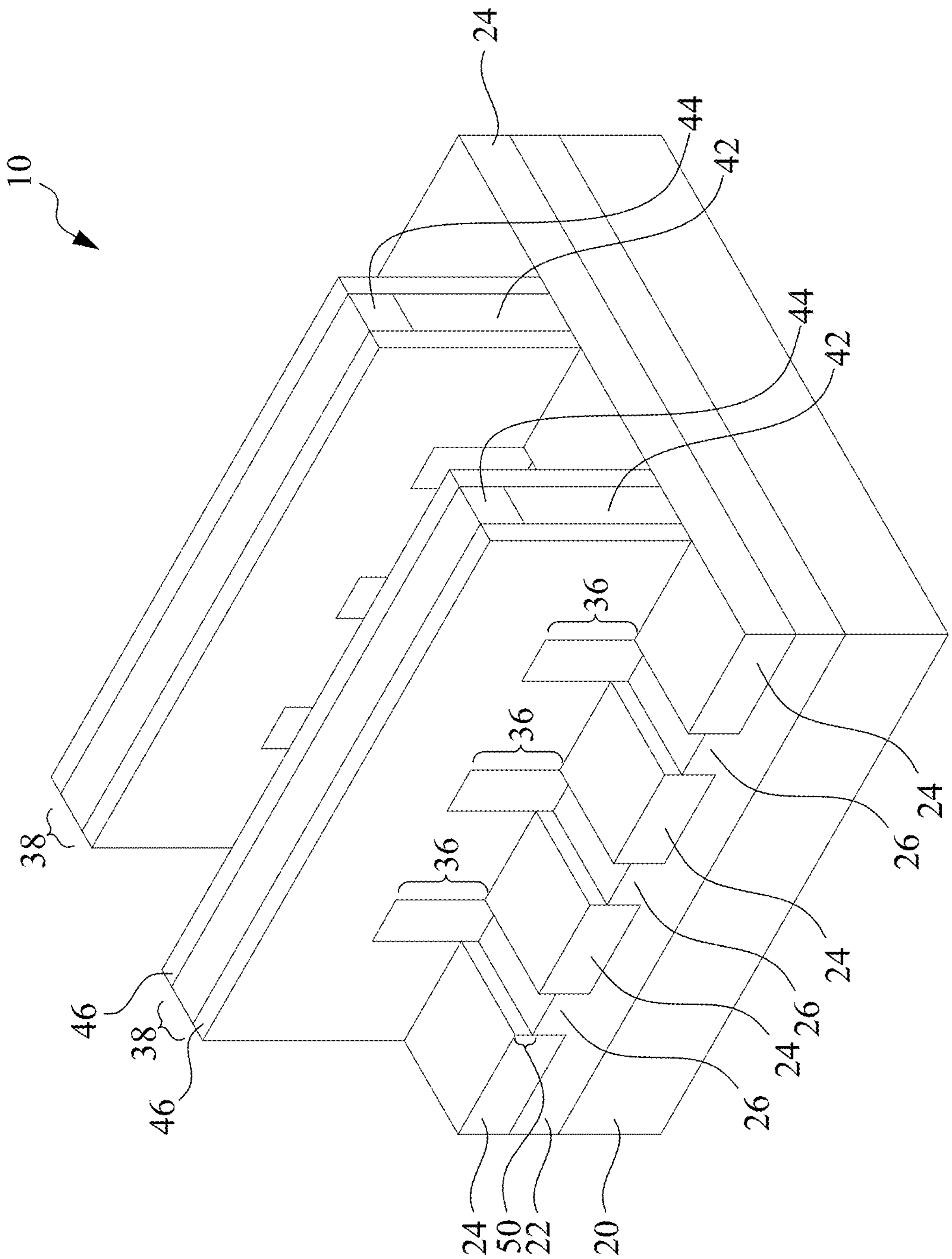
24
44
42
44
42
24A
36
24
26
24
24
26
24
26
24
24
22
20
36
46
38
46
46
38
46

FIG. 4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24
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44
42
36
36
36
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46
38
46
24
50
22
20
26
24
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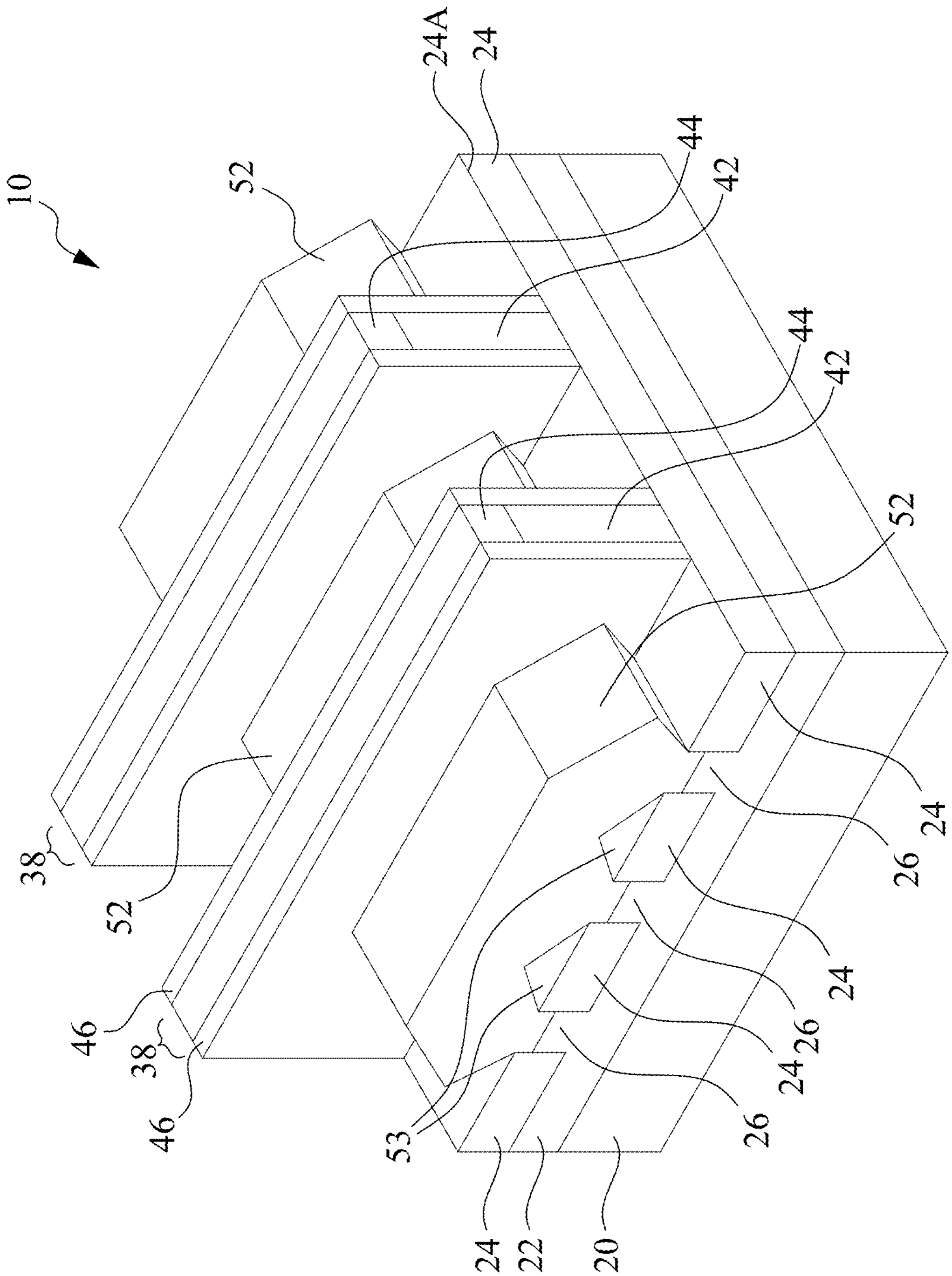
24
26
24
24
24
24

FIG. 5


10
52
24A
24
44
42
44
42
52
52
38
52
46
38
46
53
24
22
20
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24
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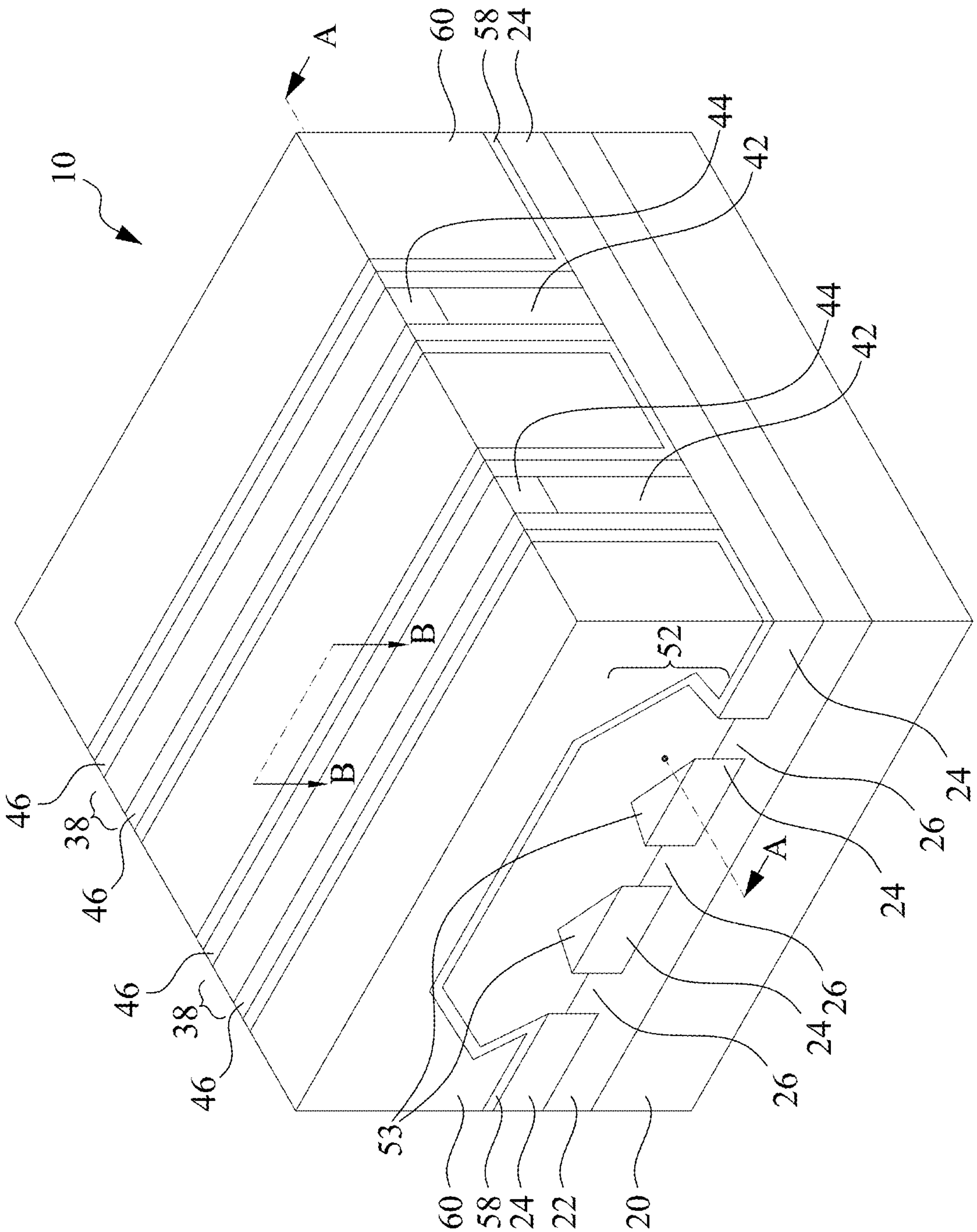
24
26
24
26
24
24

FIG. 6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A
60
58
24
44
42
44
42
B
B
52
24
26
24
A
26
24
24
26
46
38
46
46
38
46
53
60
58
24
22
20

FIG. 7A

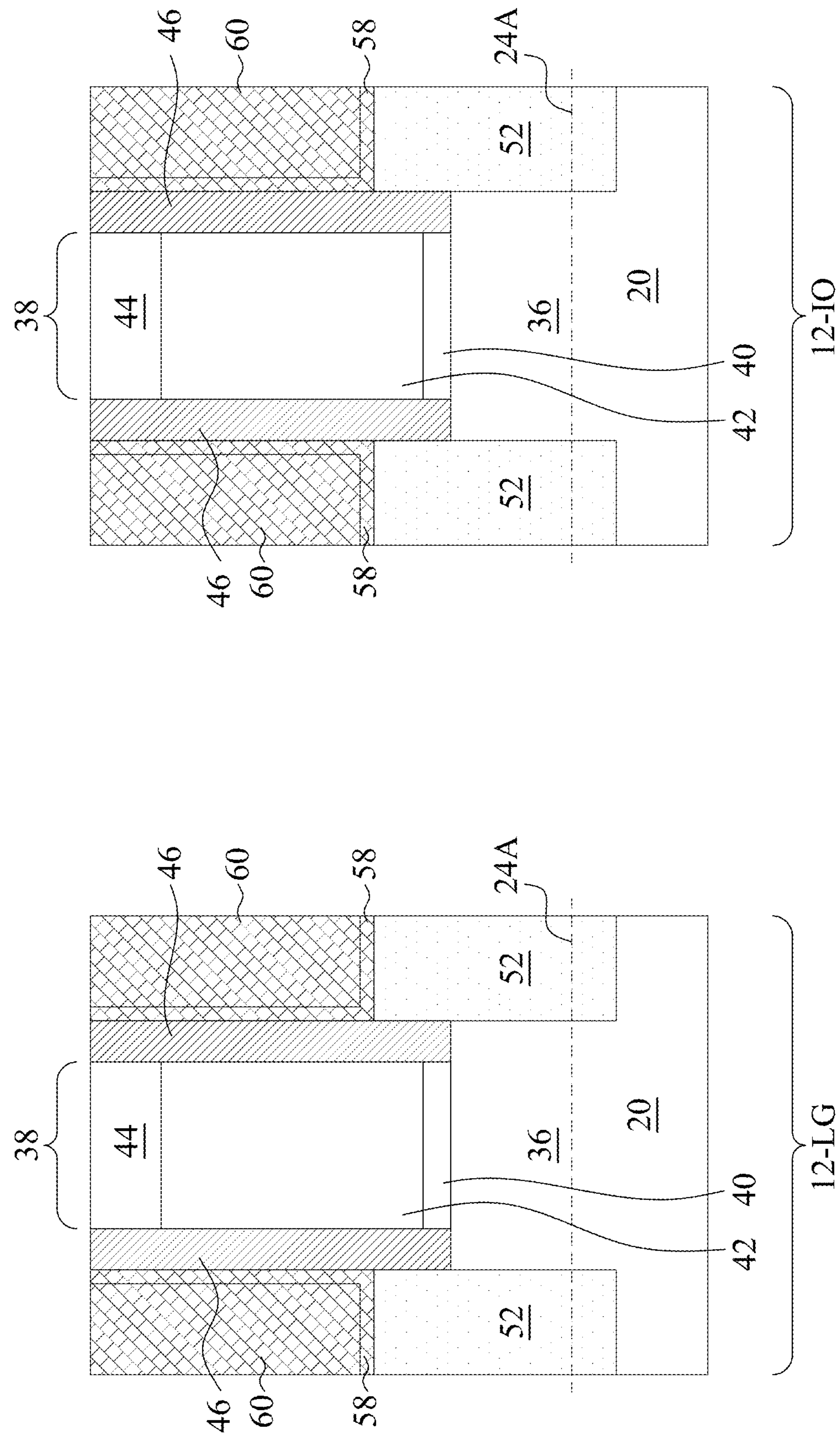

46
60
58
24A
52
38
44
36
20
40
42
52
46
60
58
12-IO
46
60
58
24A
52
38
44
36
20
40
42
52
46
60
58
12-LG

FIG. 7B

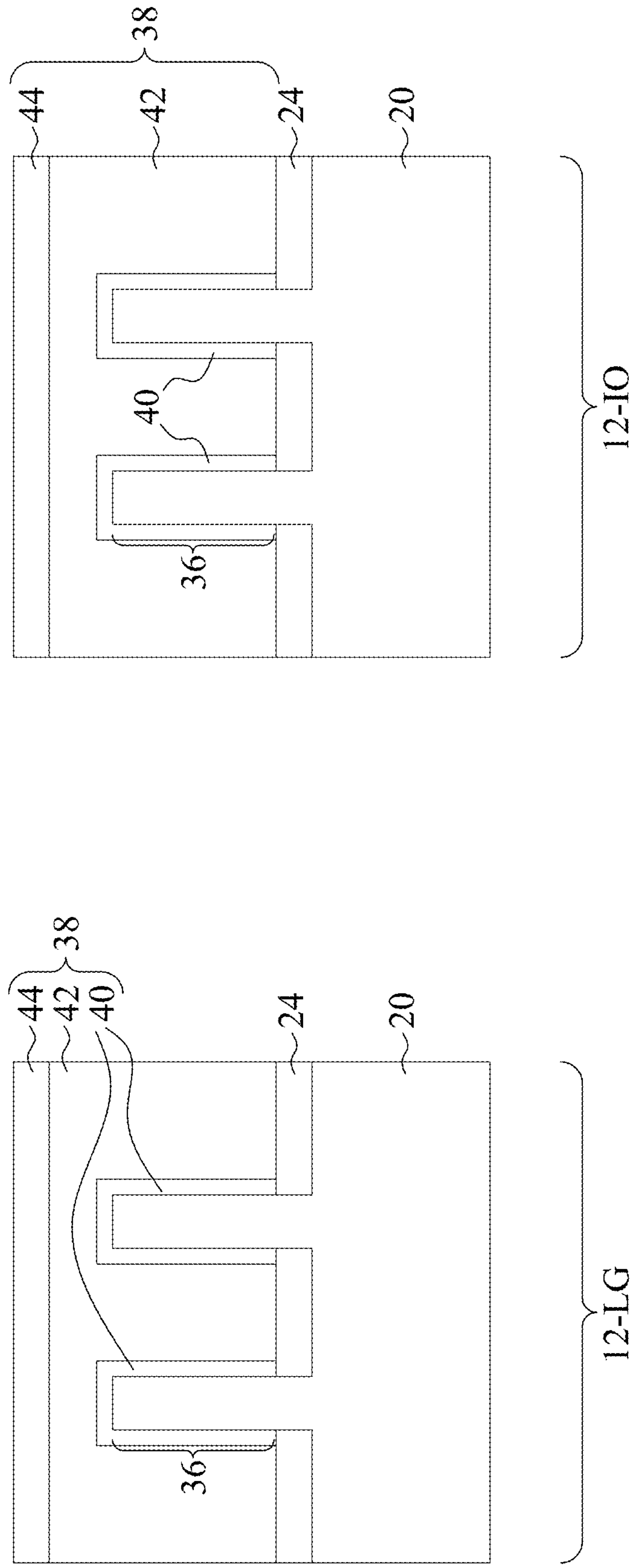


FIG. 7C

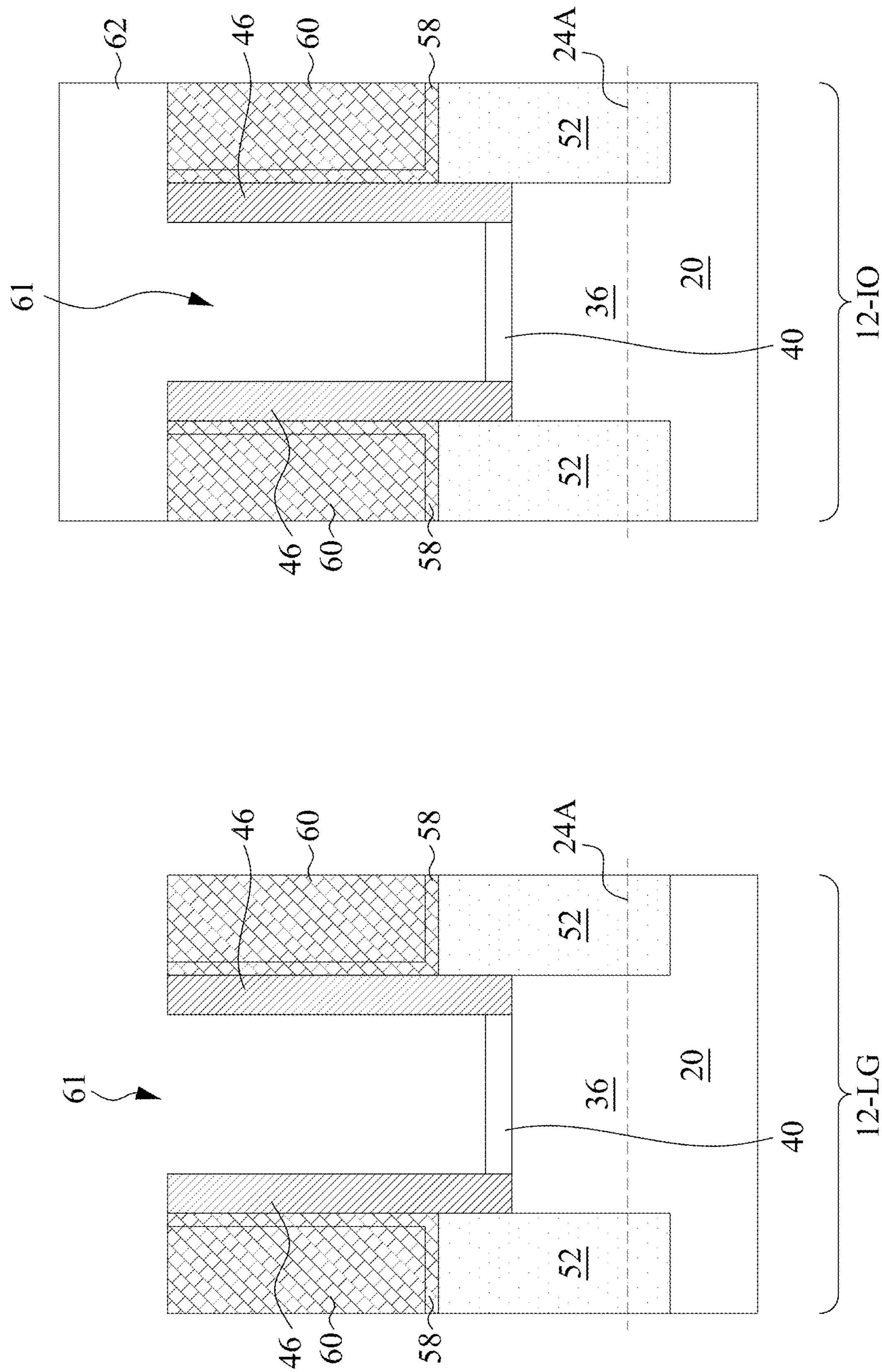


FIG. 8A

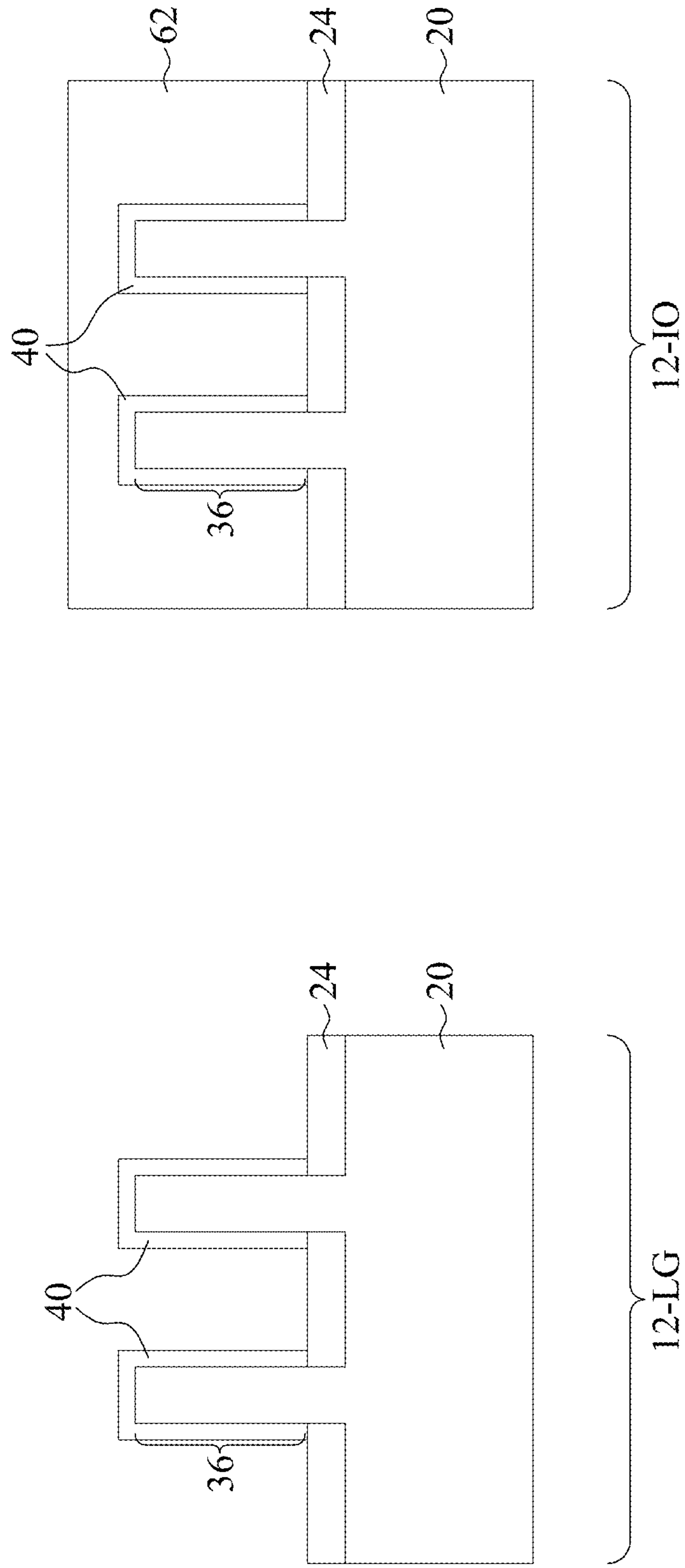


FIG. 8B

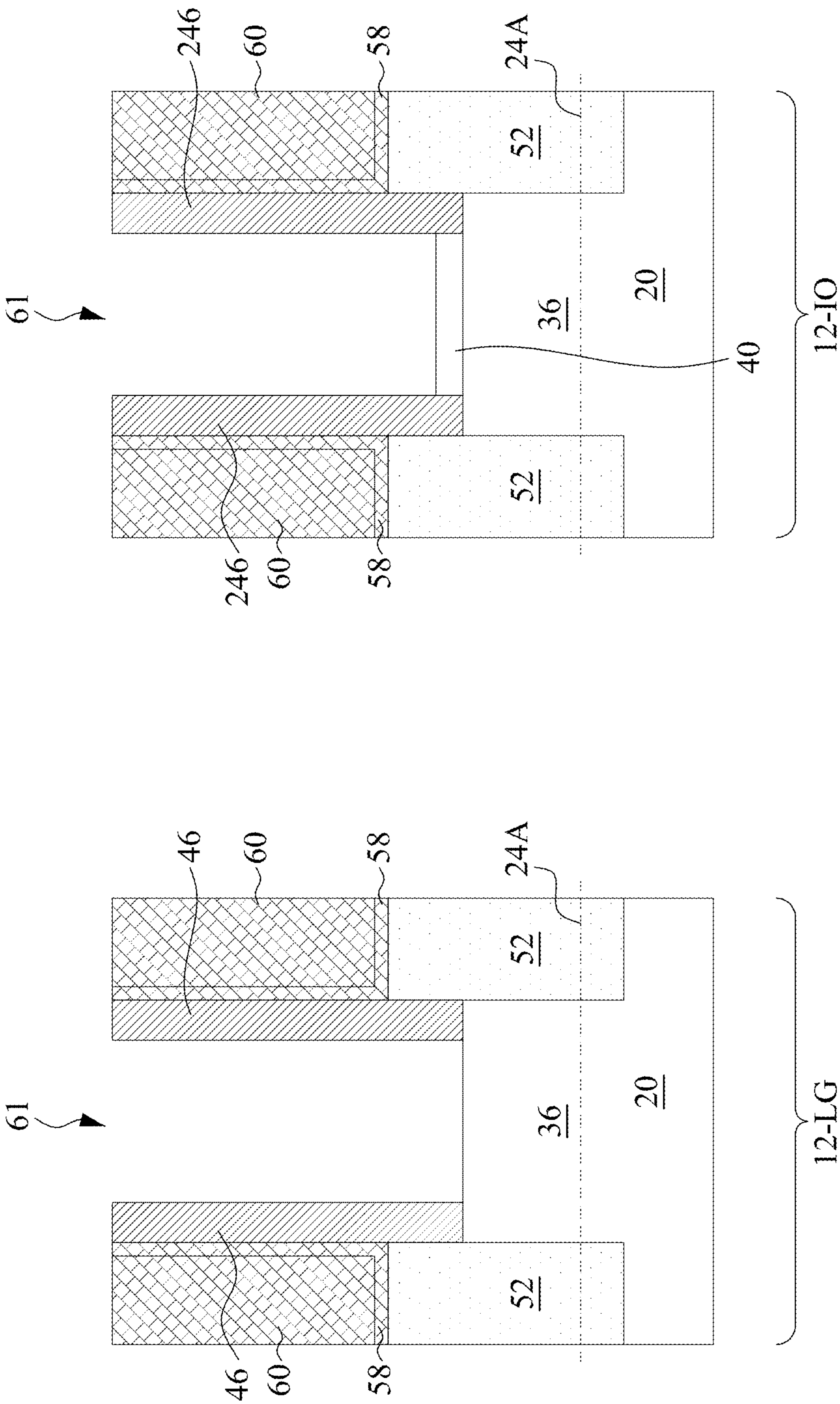

246
60
58
24A
52
61
36
20
40
12-IO
246
60
58
46
60
58
24A
52
61
36
20
12-LG
46
60
58

FIG. 9A

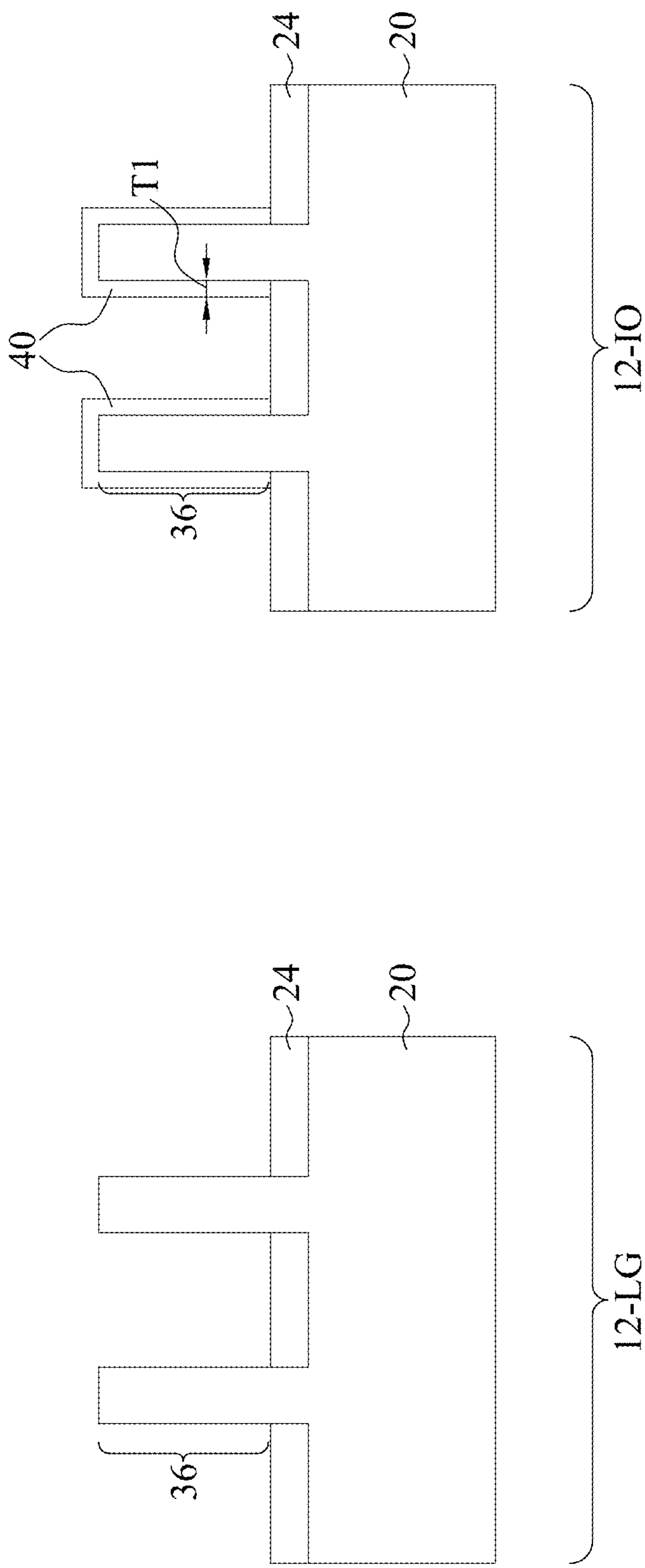


FIG. 9B

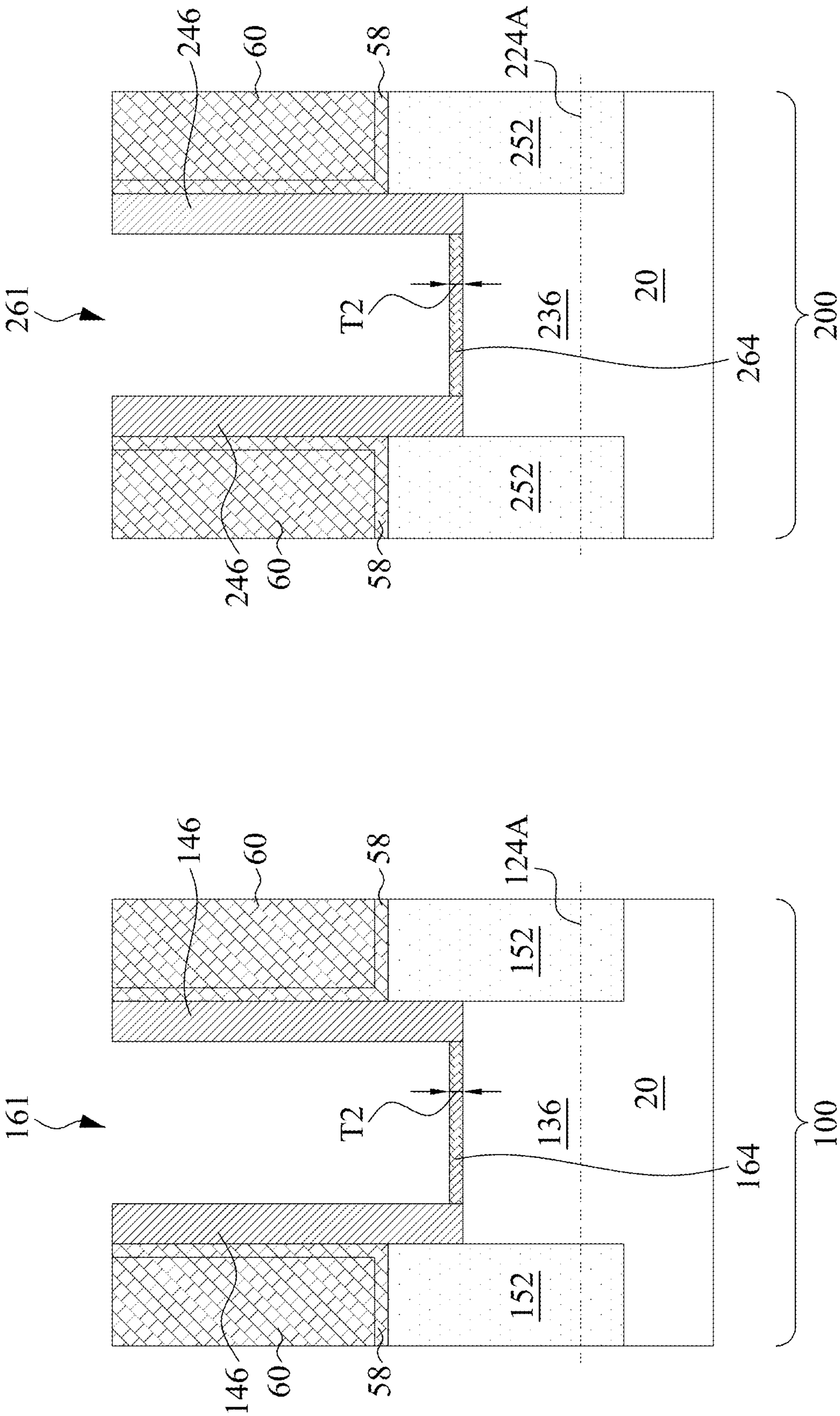

246
60
58
224A
252
261
T2
236
20
264
200
252
246
60
58
146
60
58
124A
152
161
T2
136
20
164
100
152
146
60
58

FIG. 10

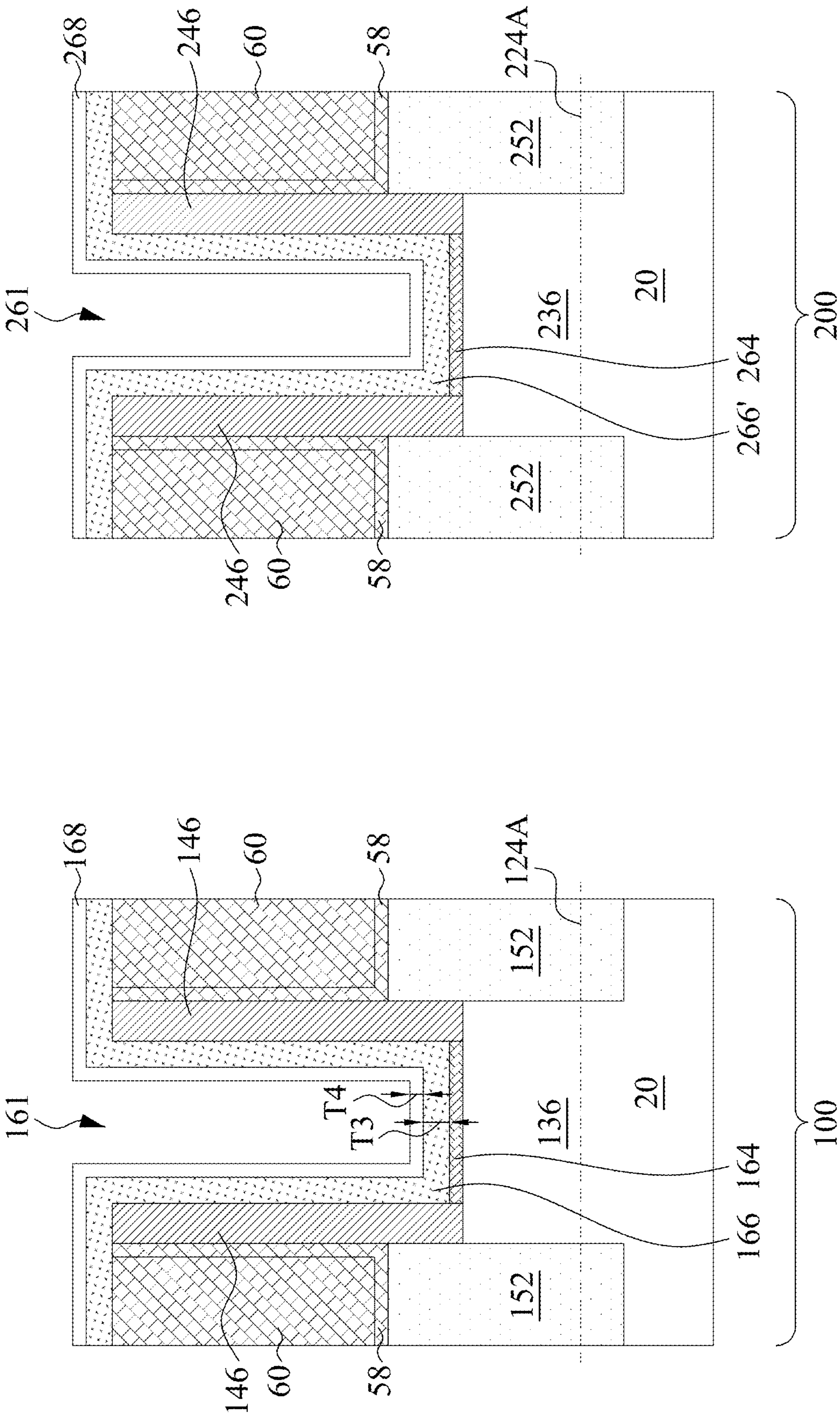

268
246
60
58
224A
252
261
236
20
264
266'
200
168
146
124A
152
161
T4
T3
136
164
166
100

FIG. 11

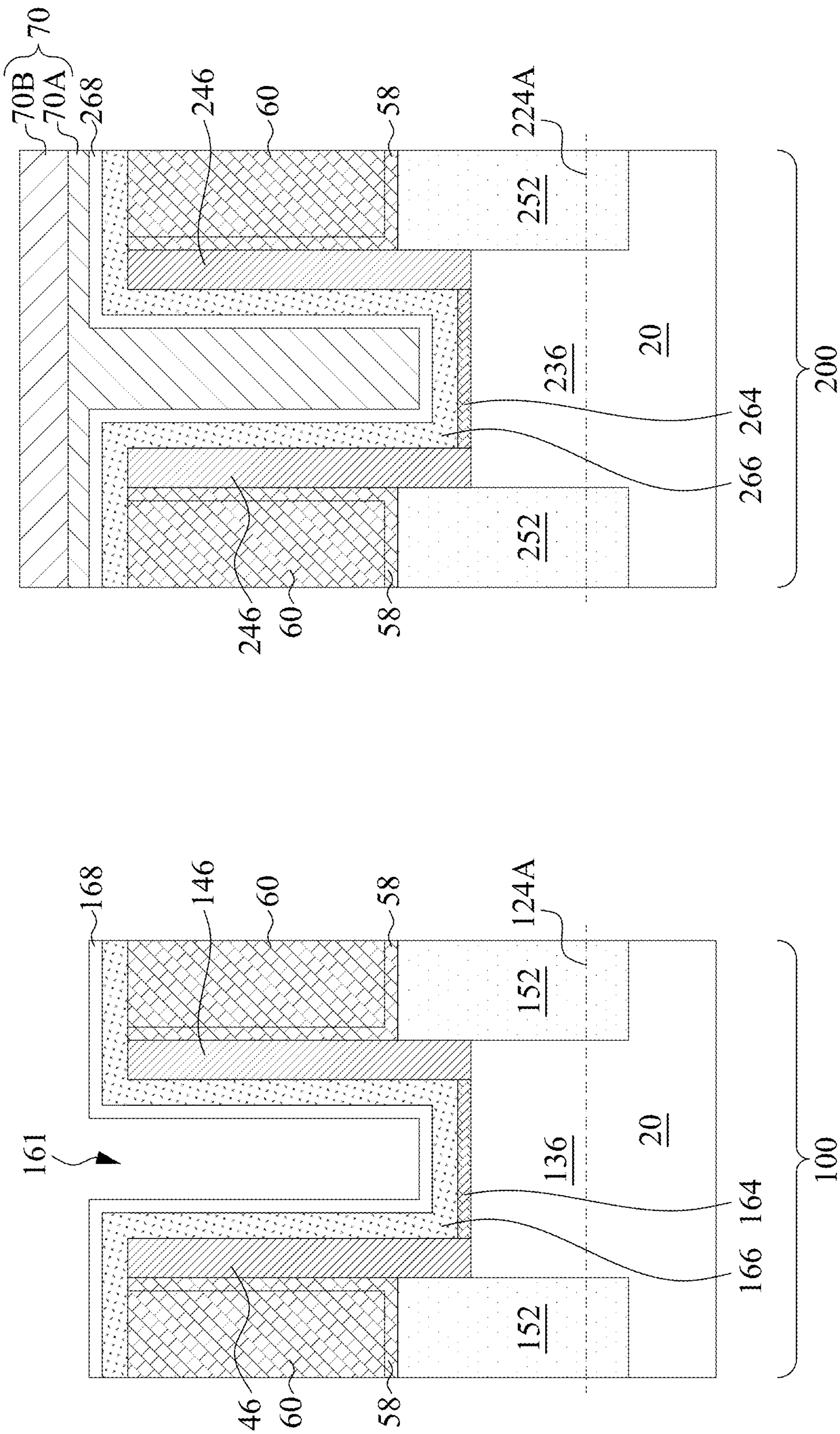

70
70B
70A
268
246
60
58
224A
252
20
236
264
266
252
246
60
58
200
168
146
60
58
124A
152
161
20
136
164
166
152
146
60
58
100

FIG. 12

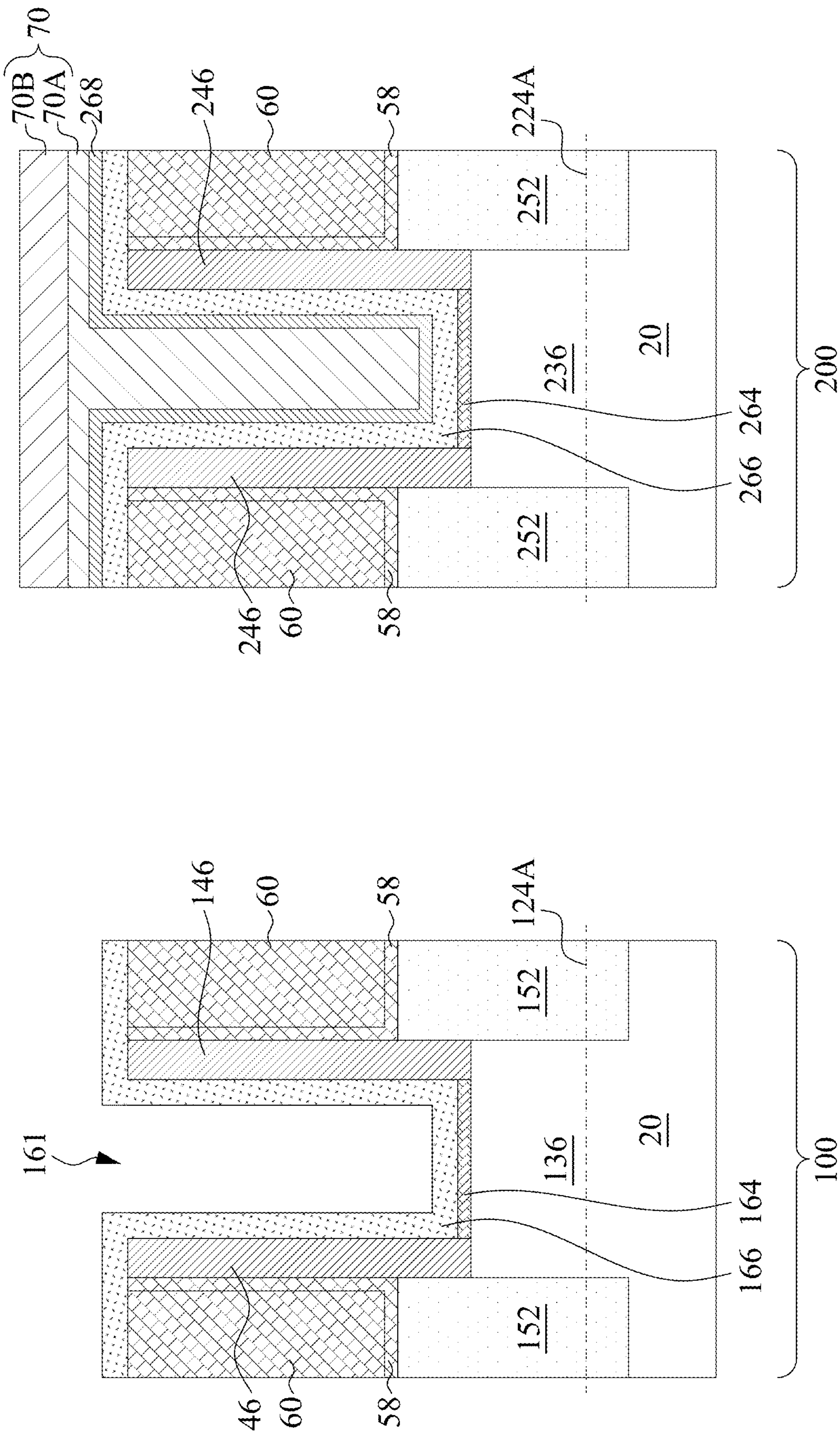
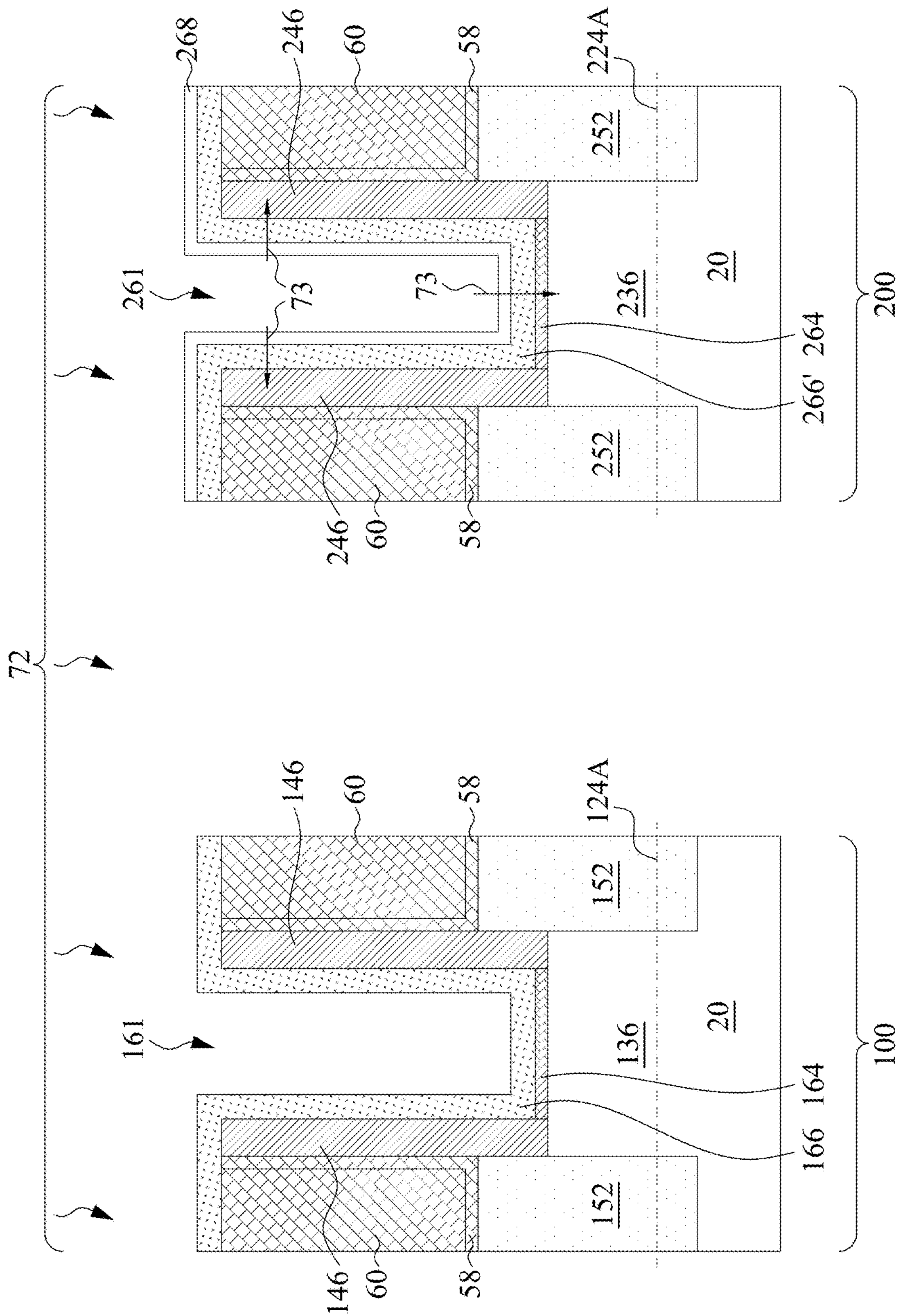

70
70B
70A
268
246
60
58
224A
252
20
236
264
266
200
246
60
58
161
146
60
58
124A
152
136
20
164
166
100
146
60
58
152

FIG. 13


72
268
246
60
58
224A
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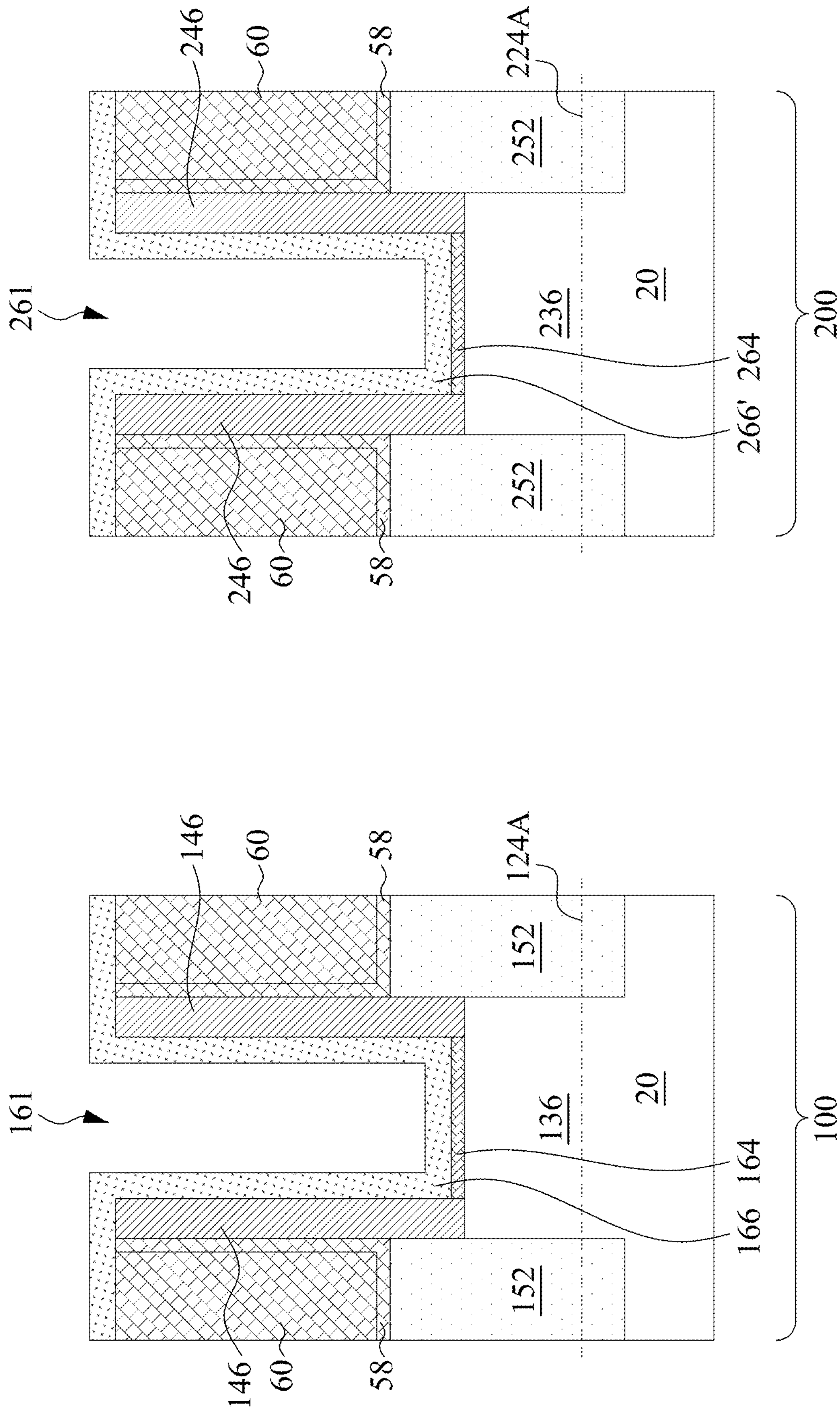
261
73
73
236
20
264
266'
246
60
58
252
200
146
60
58
124A
152
161
136
20
164
166
146
60
58
152
100

FIG. 14


246
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264
266'
252
246
60
58
200
146
60
58
124A
152
161
136
20
164
166
152
146
60
58
100

FIG. 15

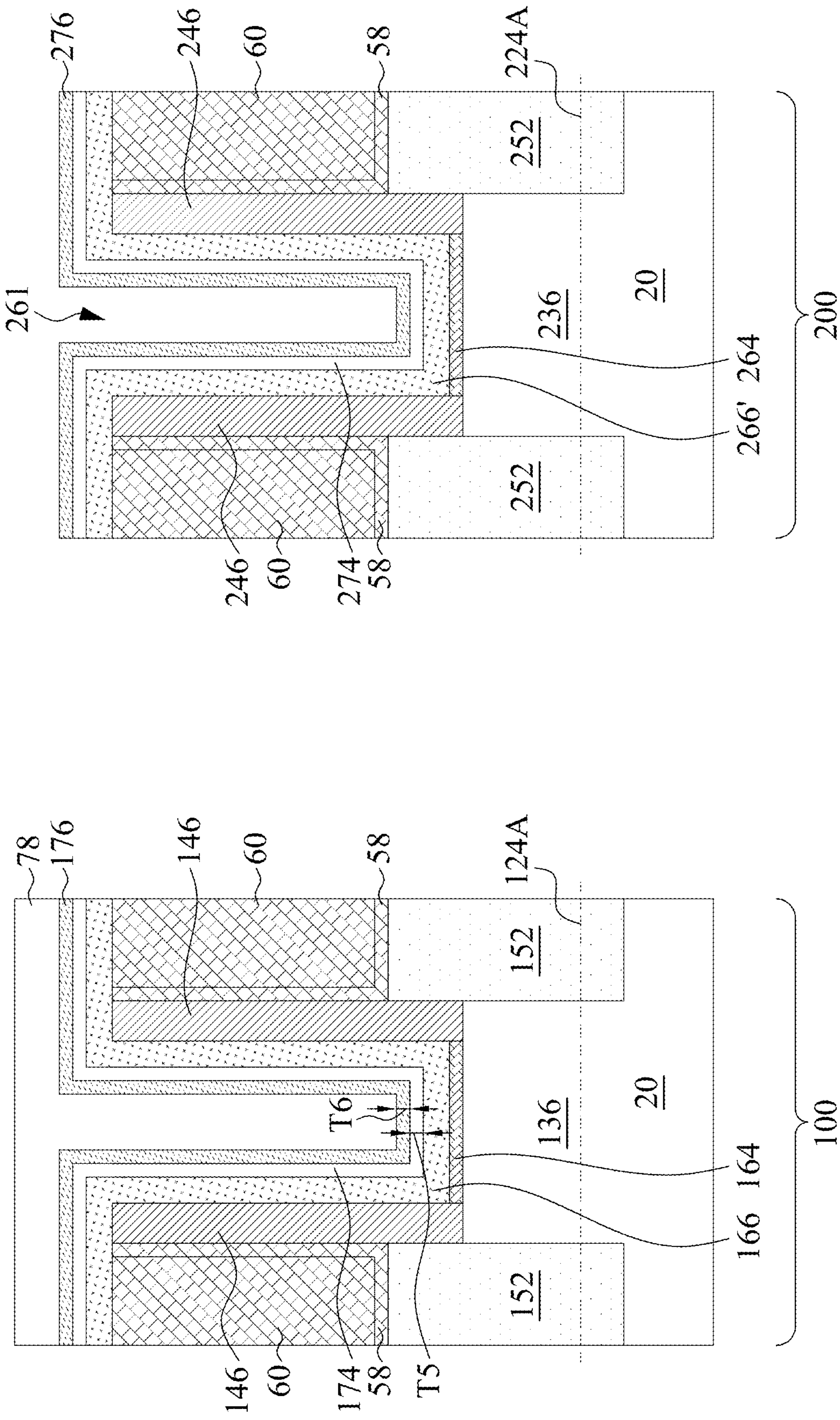
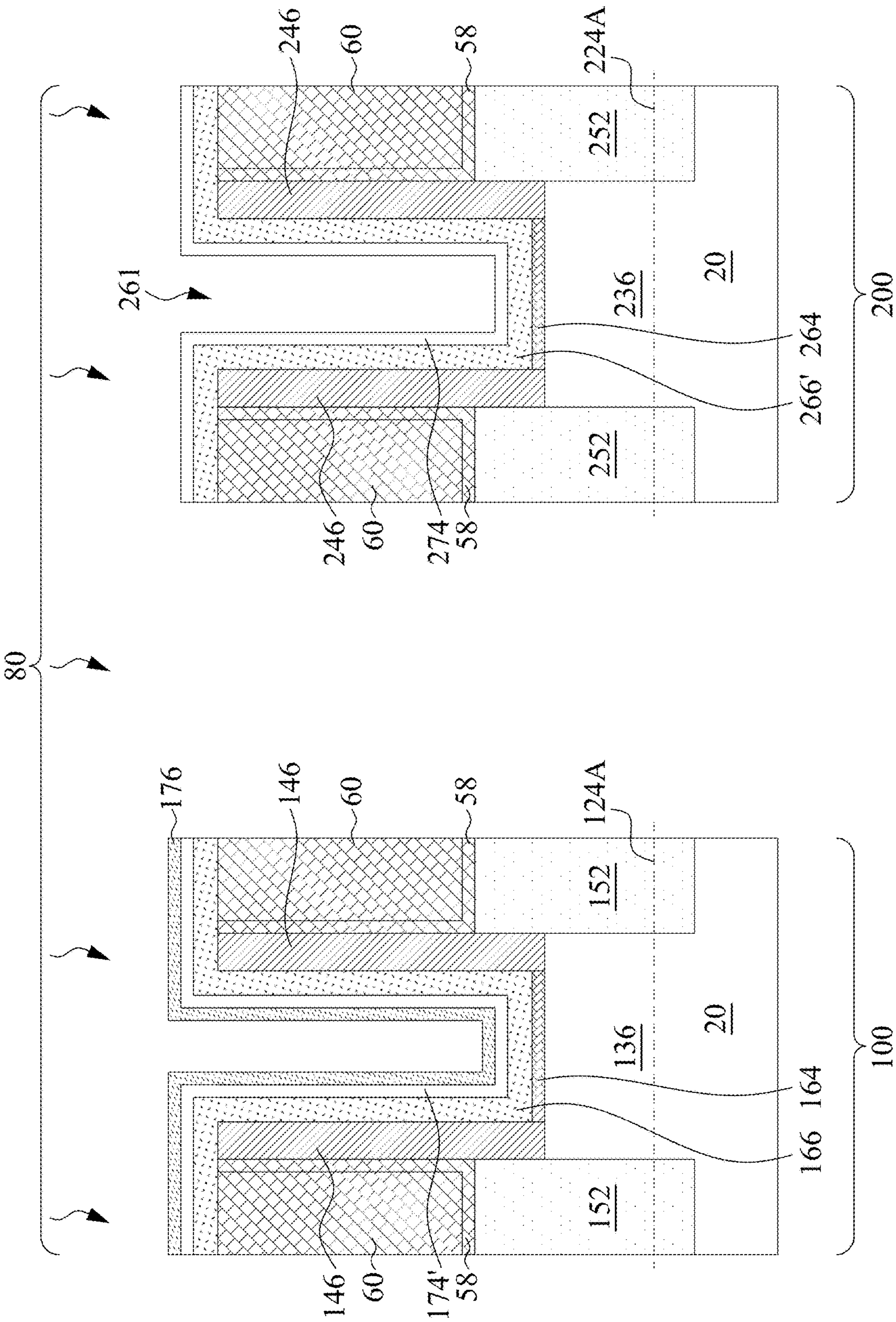

276
246
60
58
224A
252
252
261
20
236
264
266'
200
246
60
274
58
78
176
146
60
58
124A
152
152
T6
T5
136
20
164
166
100
146
60
174
58

FIG. 16


80
100
200
261
246
60
58
224A
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20
236
264
266'
274
176
146
124A
152
136
164
166
174'

FIG. 17

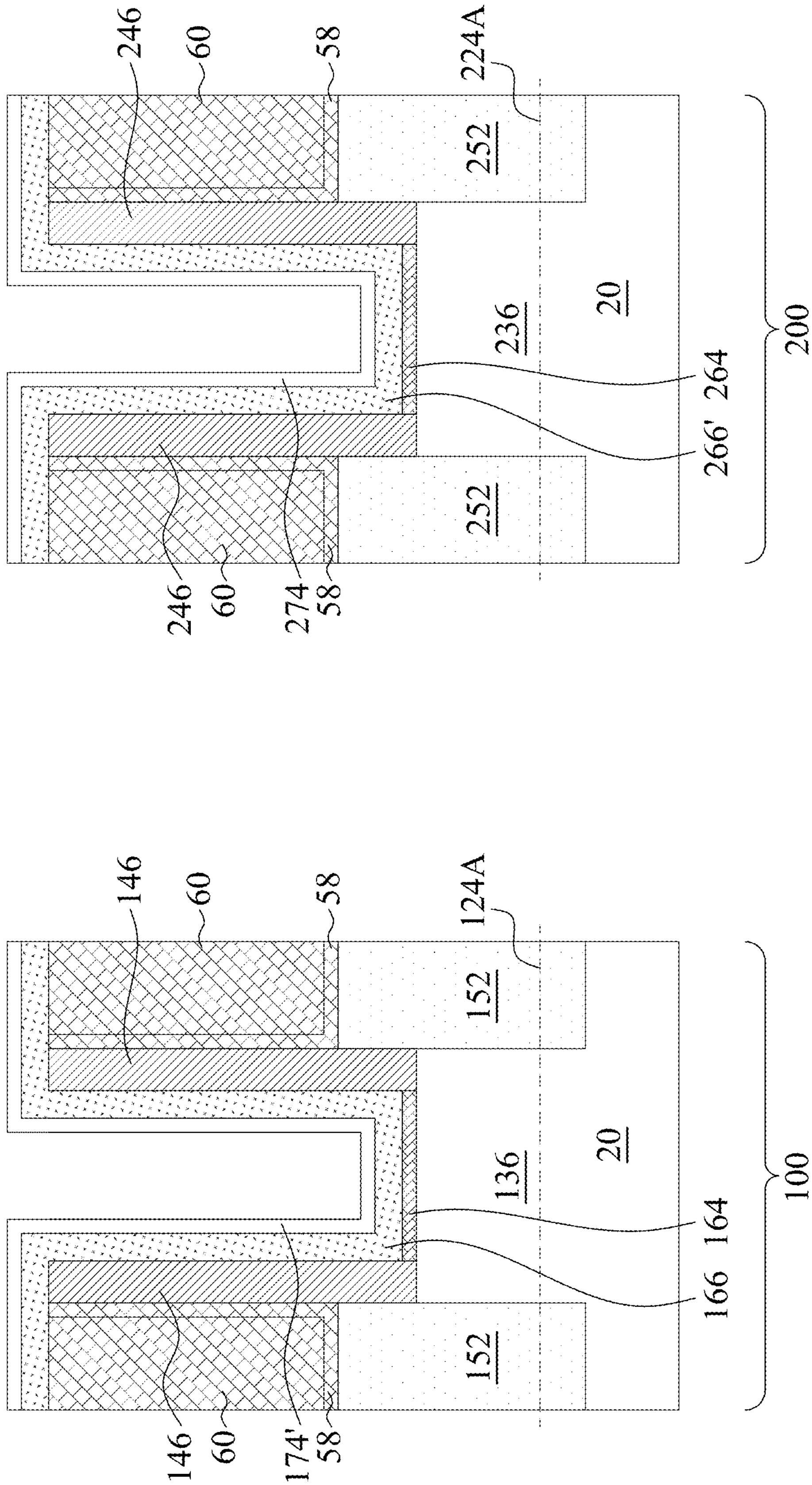

246
60
58
224A
252
236
20
264
266'
200
274
146
124A
152
136
164
166
100
174'

FIG. 18

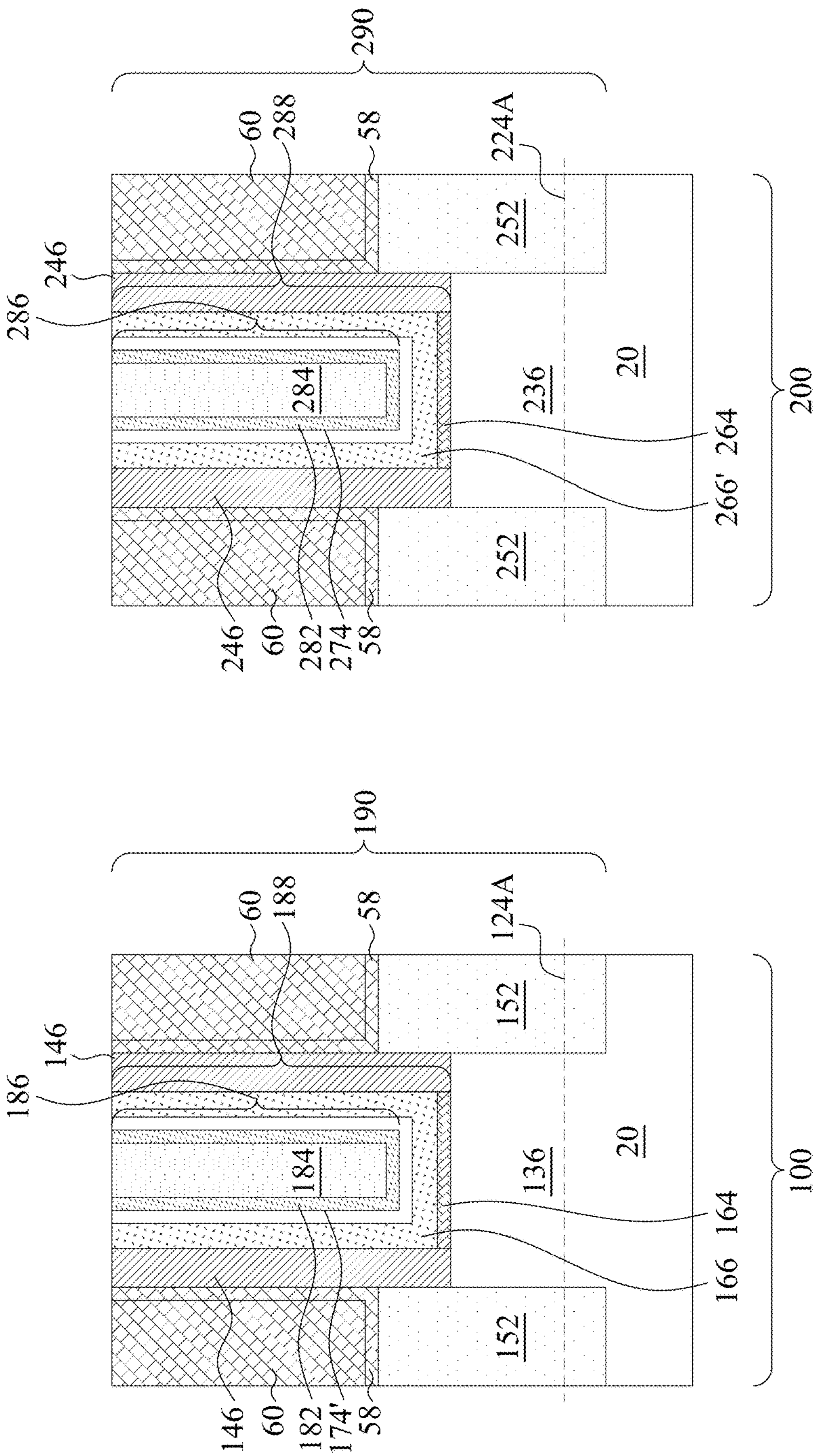
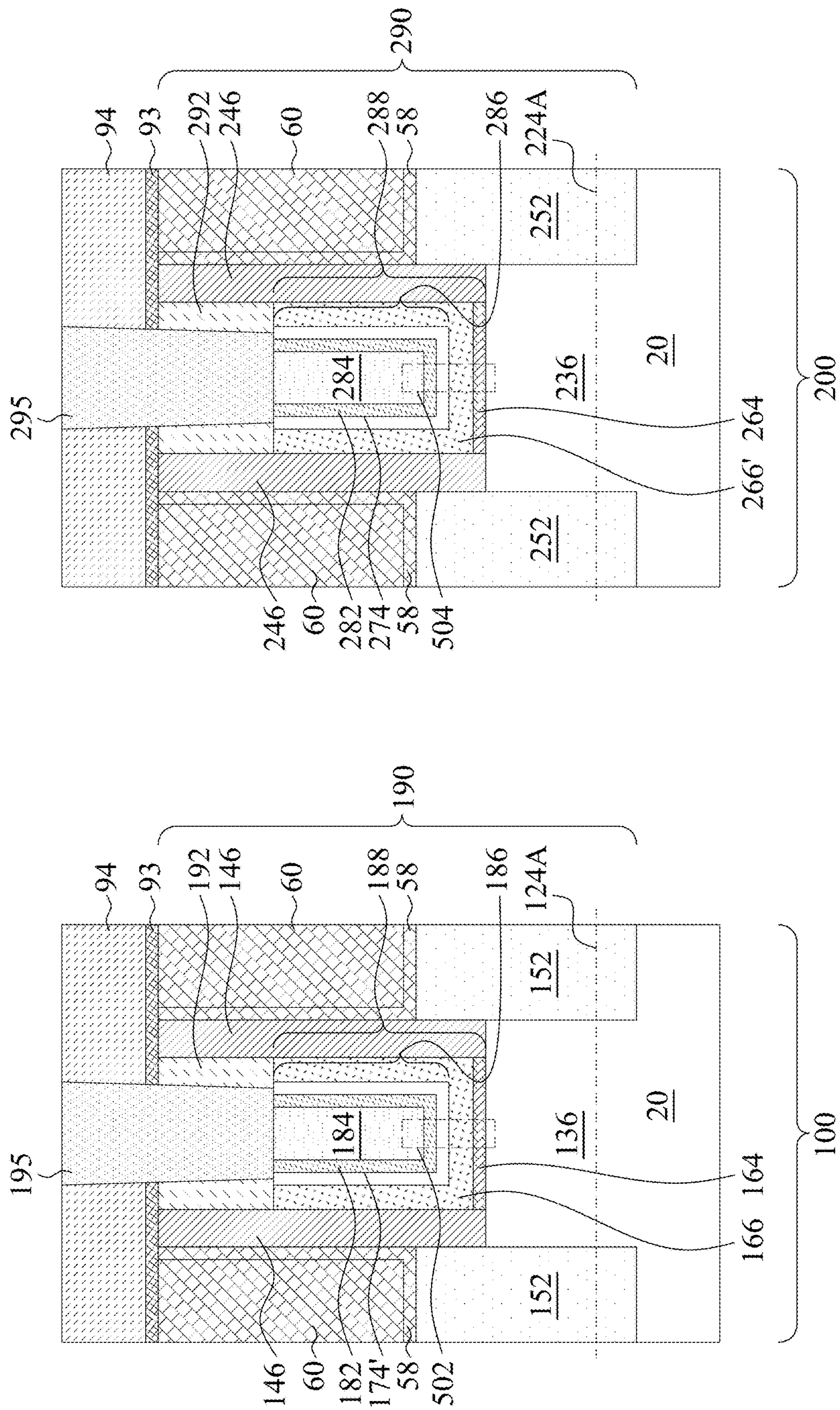

290
60
288
58
224A
246
286
252
284
236
20
264
266'
200
252
246
60
282
274
58
190
60
188
58
124A
146
186
152
184
136
20
164
166
100
152
146
60
182
174'
58

FIG. 19


290
94
93
292
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58
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224A
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284
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266'
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60
282
274
58
504
190
94
93
192
146
60
188
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186
124A
152
184
136
20
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195
164
166
152
146
60
182
174'
58
502

FIG. 20

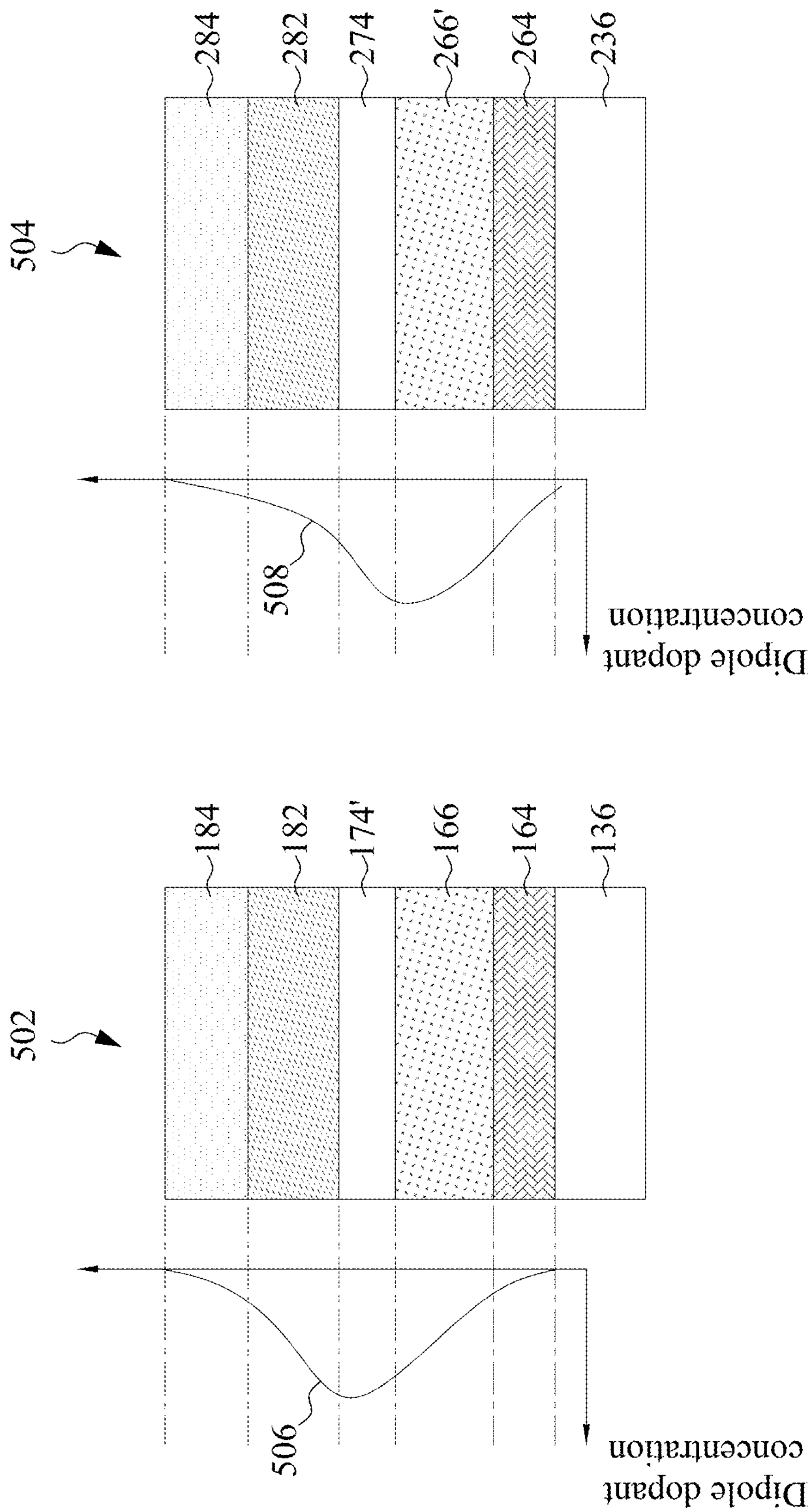
504
284
282
274
266'
264
236
508
Dipole dopant concentration
502
184
182
174'
166
164
136
506
Dipole dopant concentration

FIG. 21

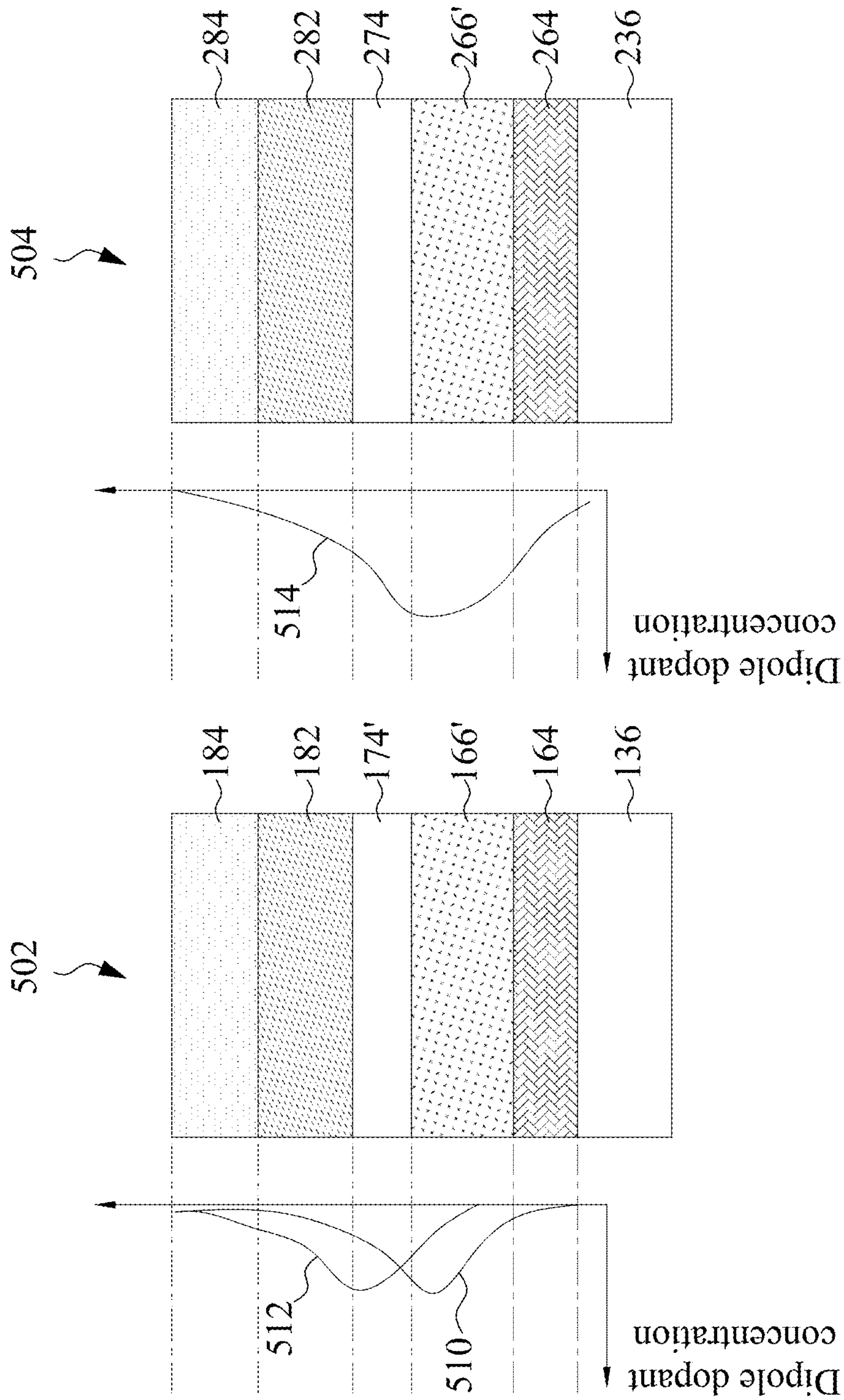
504
284
282
274
266'
264
236
514
Dipole dopant concentration
502
184
182
174'
166'
164
136
512
510
Dipole dopant concentration

FIG. 22

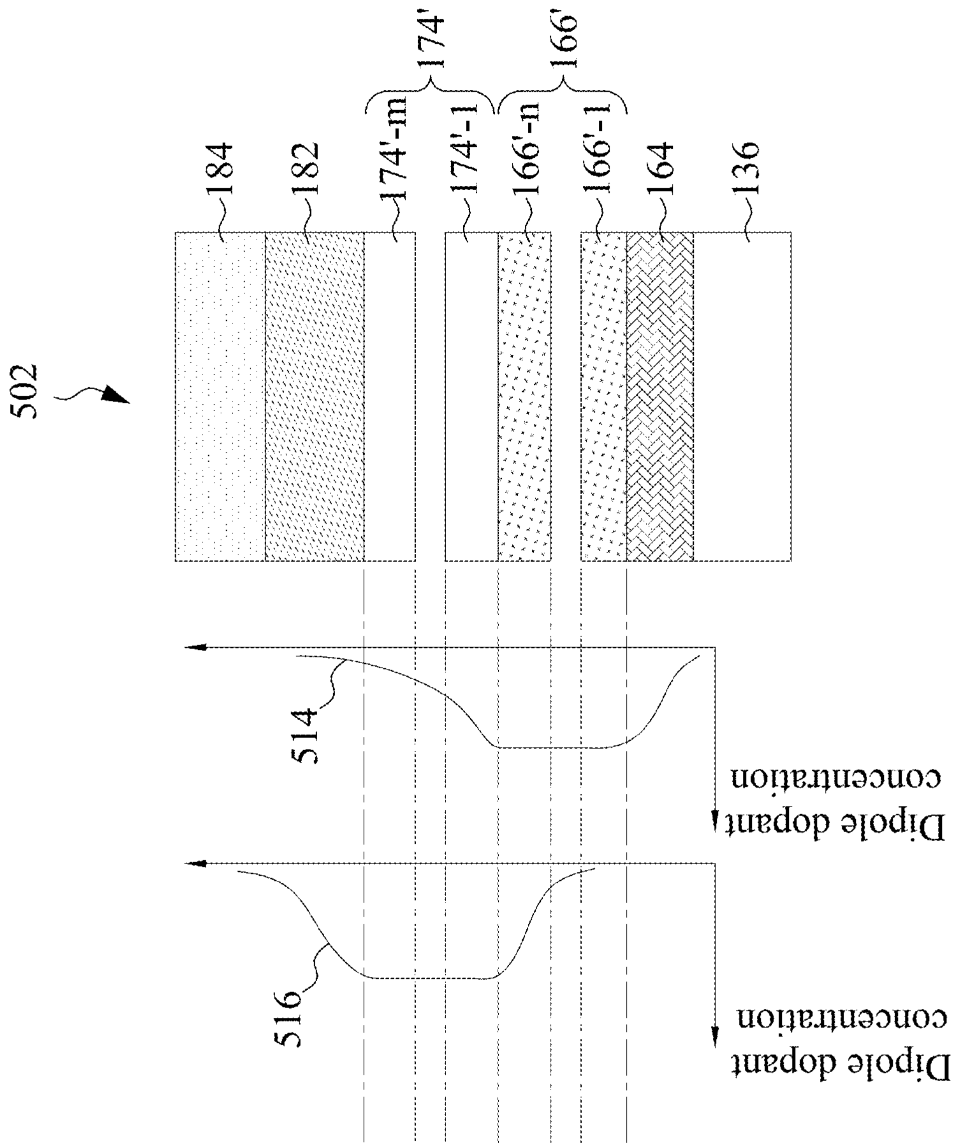
502
184
182
174'-m
174'-1
174'
166'-n
166'-1
166'
164
136
514
Dipole dopant concentration
516
Dipole dopant concentration

FIG. 23

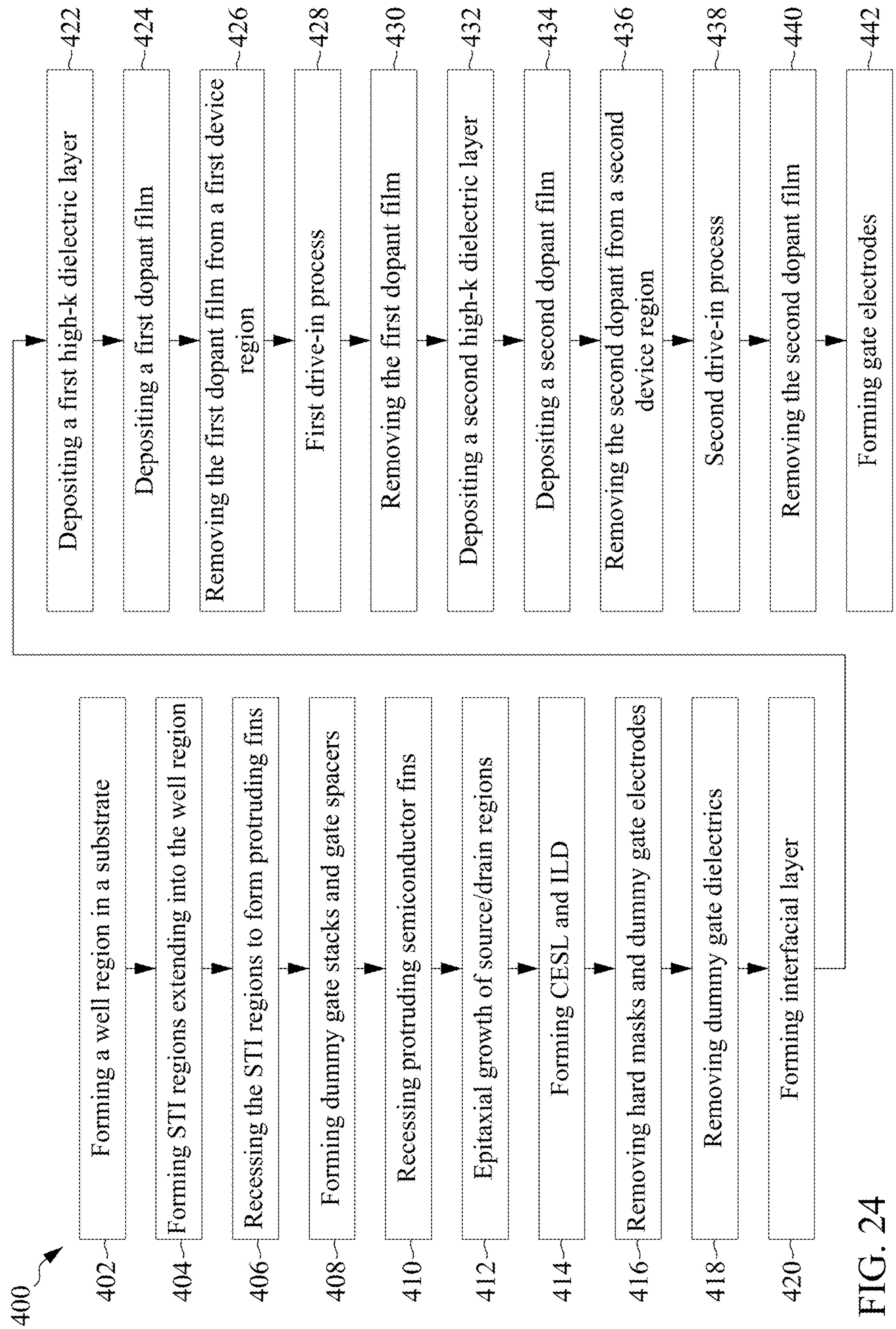
400
402 Forming a well region in a substrate
404 Forming STI regions extending into the well region
406 Recessing the STI regions to form protruding fins
408 Forming dummy gate stacks and gate spacers
410 Recessing protruding semiconductor fins
412 Epitaxial growth of source/drain regions
414 Forming CESL and ILD
416 Removing hard masks and dummy gate electrodes
418 Removing dummy gate dielectrics
420 Forming interfacial layer
422 Depositing a first high-k dielectric layer
424 Depositing a first dopant film
426 Removing the first dopant film from a first device region
428 First drive-in process
430 Removing the first dopant film
432 Depositing a second high-k dielectric layer
434 Depositing a second dopant film
436 Removing the second dopant from a second device region
438 Second drive-in process
440 Removing the second dopant film
442 Forming gate electrodes

FIG. 24

DIPOLE-ENGINEERED HIGH-K GATE DIELECTRIC AND METHOD FORMING SAME

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 18/356,860, filed Jul. 21, 2023, and entitled "Dipole-Engineered High-K Gate Dielectric and Method Forming Same," which is a divisional of U.S. patent application Ser. No. 17/094,241, filed Nov. 10, 2020, and entitled "Dipole-Engineered High-K Gate Dielectric and Method Forming Same," now U.S. Pat. No. 11,784,052, issued Oct. 10, 2023, which claims the benefit of the U.S. Provisional Application No. 63/031,099, filed on May 28, 2020, and entitled "Novel High-k Gate Oxide Stack Engineering for Device Performance Boost," which applications are hereby incorporated herein by reference.

BACKGROUND

Metal-Oxide-Semiconductor (MOS) devices are basic building elements in integrated circuits. Recent development of the MOS devices includes forming replacement gates, which include high-k gate dielectrics and metal gate electrodes over the high-k gate dielectrics. The formation of a replacement gate typically involves depositing a high-k gate dielectric layer and metal layers over the high-k gate dielectric layer, and then performing Chemical Mechanical Polish (CMP) to remove excess portions of the high-k gate dielectric layer and the metal layers. The remaining portions of the metal layers form the metal gates.

In conventional formation methods of the MOS devices, the threshold voltages of the MOS devices may be adjusted by performing a thermal anneal process when conducting ammonia to treat the high-k dielectric layers. Although the threshold voltage can be changed, it was difficult to adjust the threshold voltages to intended values, and further adjustment had to be achieved by adopting different work-function metals and adjusting the thickness of the work-function metals.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. **1-6**, **7**A, **7**B, **7**C, **8**A, **8**B, **9**A, **9**B, and **10-20** illustrate the perspective views and cross-sectional views of intermediate stages in the formation of Fin Field-Effect Transistors (FinFETs) in accordance with some embodiments.

FIGS. **21** through **23** illustrate the distributions of dipole dopants in accordance with some embodiments.

FIG. **24** illustrates a process flow for forming FinFETs in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "underlying," "below," "lower," "overlying," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Transistors with dipole-engineered high-k dielectric layers and the method of incorporating the dipole dopants into the high-k dielectric layers are provided in accordance with various embodiments. Dipole dopants are diffused into the high-k dielectric layers through thermal diffusion. The threshold voltages of the corresponding transistors are adjusted. The magnitude of the adjustment depends from the material of high-k dielectric layer and the position of the doping. Accordingly, more than one high-k dielectric layers are formed, which may have different dielectric constant (k) values. The dipole dopants may be selectively doped into one or more of the high-k dielectric layers to provide different threshold-voltage adjustment ability. Furthermore, device performance is improved through doping the dipole dopants. The Capacitance Equivalent Thickness (CET) of the high-k dielectric layers is reduced. The intermediate stages of forming the transistors are illustrated in accordance with some embodiments. Some variations of some embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. In accordance with some embodiments, the formation of Fin Field-Effect Transistors (FinFETs) is used as an example to explain the concept of the present disclosure. Other types of transistors such as planar transistors and Gate-All-Around (GAA) transistors may also adopt the concept of the present disclosure.

FIGS. **1-6**, **7**A, **7**B, **7**C, **8**A, **8**B, **9**A, **9**B, and **10-20** illustrate the cross-sectional views and perspective views of intermediate stages in the formation of Fin Field-Effect Transistors (FinFETs) in accordance with some embodiments of the present disclosure. The processes shown in these figures are also reflected schematically in the process flow **400** as shown in FIG. **24**.

In FIG. **1**, substrate **20** is provided. The substrate **20** may be a semiconductor substrate, such as a bulk semiconductor substrate, a Semiconductor-On-Insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The semiconductor substrate **20** may be a part of wafer **10**, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a Buried Oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon substrate or a glass substrate. Other substrates such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of semiconductor substrate **20** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

Further referring to FIG. **1**, well region **22** is formed in substrate **20**. The respective process is illustrated as process **402** in the process flow **400** as shown in FIG. **24**. In accordance with some embodiments of the present disclosure, well region **22** is an n-type well region formed through implanting an n-type impurity, which may be phosphorus, arsenic, antimony, or the like, into substrate **20**. In accordance with other embodiments of the present disclosure, well region **22** is a p-type well region formed through implanting a p-type impurity, which may be boron, indium, or the like, into substrate **20**. The resulting well region **22** may extend to the top surface of substrate **20**. The n-type or p-type impurity concentration may be equal to or less than 10^{18} cm^{-3}, such as in the range between about 10^{17} cm^{-3} and about 10^{18} cm^{-3}.

Referring to FIG. **2**, isolation regions **24** are formed to extend from a top surface of substrate **20** into substrate **20**. Isolation regions **24** are alternatively referred to as Shallow Trench Isolation (STI) regions hereinafter. The respective process is illustrated as process **404** in the process flow **400** as shown in FIG. **24**. The portions of substrate **20** between neighboring STI regions **24** are referred to as semiconductor strips **26**. To form STI regions **24**, pad oxide layer **28** and hard mask layer **30** are formed on semiconductor substrate **20**, and are then patterned. Pad oxide layer **28** may be a thin film formed of silicon oxide. In accordance with some embodiments of the present disclosure, pad oxide layer **28** is formed in a thermal oxidation process, wherein a top surface layer of semiconductor substrate **20** is oxidized. Pad oxide layer **28** acts as an adhesion layer between semiconductor substrate **20** and hard mask layer **30**. Pad oxide layer **28** may also act as an etch stop layer for etching hard mask layer **30**. In accordance with some embodiments of the present disclosure, hard mask layer **30** is formed of silicon nitride, for example, using Low-Pressure Chemical Vapor Deposition (LPCVD). In accordance with other embodiments of the present disclosure, hard mask layer **30** is formed by thermal nitriding of silicon, or Plasma Enhanced Chemical Vapor Deposition (PECVD). A photo resist (not shown) is formed on hard mask layer **30** and is then patterned. Hard mask layer **30** is then patterned using the patterned photo resist as an etching mask to form hard masks **30** as shown in FIG. **2**.

Next, the patterned hard mask layer **30** is used as an etching mask to etch pad oxide layer **28** and substrate **20**, followed by filling the resulting trenches in substrate **20** with a dielectric material(s). A planarization process such as a Chemical Mechanical Polish (CMP) process or a mechanical grinding process is performed to remove excess portions of the dielectric materials, and the remaining portions of the dielectric materials(s) are STI regions **24**. STI regions **24** may include a liner dielectric (not shown), which may be a thermal oxide formed through a thermal oxidation of a surface layer of substrate **20**. The liner dielectric may also be a deposited silicon oxide layer, silicon nitride layer, or the like formed using, for example, Atomic Layer Deposition (ALD), High-Density Plasma Chemical Vapor Deposition (HDPCVD), or Chemical Vapor Deposition (CVD). STI regions **24** may also include a dielectric material over the liner oxide, wherein the dielectric material may be formed using Flowable Chemical Vapor Deposition (FCVD), spin-on coating, or the like. The dielectric material over the liner dielectric may include silicon oxide in accordance with some embodiments.

The top surfaces of hard masks **30** and the top surfaces of STI regions **24** may be substantially level with each other. Semiconductor strips **26** are between neighboring STI regions **24**. In accordance with some embodiments of the present disclosure, semiconductor strips **26** are parts of the original substrate **20**, and hence the material of semiconductor strips **26** is the same as that of substrate **20**. In accordance with alternative embodiments of the present disclosure, semiconductor strips **26** are replacement strips formed by etching the portions of substrate **20** between STI regions **24** to form recesses, and performing an epitaxy to regrow another semiconductor material in the recesses. Accordingly, semiconductor strips **26** are formed of a semiconductor material different from that of substrate **20**. In accordance with some embodiments, semiconductor strips **26** are formed of silicon germanium, silicon carbon, or a III-V compound semiconductor material.

Referring to FIG. **3**, STI regions **24** are recessed, so that the top portions of semiconductor strips **26** protrude higher than the top surfaces **24**A of the remaining portions of STI regions **24** to form protruding fins **36**. The respective process is illustrated as process **406** in the process flow **400** as shown in FIG. **24**. The etching may be performed using a dry etching process, wherein the mixture of HF and NH_3, for example, is used as the etching gas. During the etching process, plasma may be generated. Argon may also be included. In accordance with alternative embodiments of the present disclosure, the recessing of STI regions **24** is performed using a wet etch process. The etching chemical may include HF, for example.

In above-illustrated embodiments, the fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins.

Referring to FIG. **4**, dummy gate stacks **38** are formed to extend on the top surfaces and the sidewalls of (protruding) fins **36**. The respective process is illustrated as process **408** in the process flow **400** as shown in FIG. **24**. Dummy gate stacks **38** may include dummy gate dielectrics **40** (shown in FIGS. **7**B and **7**C) and dummy gate electrodes **42** over dummy gate dielectrics **40**. Dummy gate electrodes **42** may be formed, for example, using polysilicon or amorphous silicon, and other materials may also be used. Each of dummy gate stacks **38** may also include one (or a plurality of) hard mask layer **44** over dummy gate electrodes **42**. Hard mask layers **44** may be formed of silicon nitride, silicon oxide, silicon carbo-nitride, or multi-layers thereof. Dummy gate stacks **38** may cross over a single one or a plurality of protruding fins **36** and/or STI regions **24**. Dummy gate stacks **38** also have lengthwise directions perpendicular to the lengthwise directions of protruding fins **36**.

Next, gate spacers **46** are formed on the sidewalls of dummy gate stacks **38**. The respective process is also shown as process **408** in the process flow **400** as shown in FIG. **24**. In accordance with some embodiments of the present disclosure, gate spacers **46** are formed of a dielectric material(s) such as silicon nitride, silicon carbo-nitride, or the like, and may have a single-layer structure or a multi-layer structure including a plurality of dielectric layers.

The portions of protruding fins **36** that are not covered by dummy gate stacks **38** and gate spacers **46** are then etched, resulting in the structure shown in FIG. **5**. The respective process is illustrated as process **410** in the process flow **400** as shown in FIG. **24**. The recessing may be anisotropic, and hence the portions of fins **36** directly underlying dummy gate stacks **38** and gate spacers **46** are protected, and are not etched. The top surfaces of the recessed semiconductor strips **26** may be lower than the top surfaces **24**A of STI regions **24** in accordance with some embodiments. Recesses **50** are accordingly formed. Recesses **50** comprise portions located on the opposite sides of dummy gate stacks **38**, and portions between remaining portions of protruding fins **36**.

Next, epitaxy regions (source/drain regions) **52** are formed by selectively growing (through epitaxy) a semiconductor material in recesses **50**, resulting in the structure in FIG. **6**. The respective process is illustrated as process **412** in the process flow **400** as shown in FIG. **24**. Depending on whether the resulting FinFET is a p-type FinFET or an n-type FinFET, a p-type or an n-type impurity may be in-situ doped with the proceeding of the epitaxy. For example, when the resulting FinFET is a p-type FinFET, silicon germanium boron (SiGeB), silicon boron (SiB), or the like may be grown. Conversely, when the resulting FinFET is an n-type FinFET, silicon phosphorous (SiP), silicon carbon phosphorous (SiCP), or the like may be grown. In accordance with alternative embodiments of the present disclosure, epitaxy regions **52** comprise III-V compound semiconductors such as GaAs, InP, GaN, InGaAs, InAlAs, GaSb, AlSb, AlAs, AlP, GaP, combinations thereof, or multi-layers thereof. After Recesses **50** are filled with epitaxy regions **52**, the further epitaxial growth of epitaxy regions **52** causes epitaxy regions **52** to expand horizontally, and facets may be formed. The further growth of epitaxy regions **52** may also cause neighboring epitaxy regions **52** to merge with each other. Voids (air gaps) **53** may be generated.

After the epitaxy process, epitaxy regions **52** may be further implanted with a p-type or an n-type impurity to form source and drain regions, which are also denoted using reference numeral **52**. In accordance with alternative embodiments of the present disclosure, the implantation process is skipped when epitaxy regions **52** are in-situ doped with the p-type or n-type impurity during the epitaxy.

FIG. **7**A illustrates a perspective view of the structure after the formation of Contact Etch Stop Layer (CESL) **58** and Inter-Layer Dielectric (ILD) **60**. The respective process is illustrated as process **414** in the process flow **400** as shown in FIG. **24**. CESL **58** may be formed of silicon oxide, silicon nitride, silicon carbo-nitride, or the like, and may be formed using CVD, ALD, or the like. ILD **60** may include a dielectric material formed using, for example, FCVD, spin-on coating, CVD, or another deposition method. ILD **60** may be formed of an oxygen-containing dielectric material, which may be a silicon-oxide based material formed using Tetra Ethyl Ortho Silicate (TEOS) as a precursor, Phospho-Silicate Glass (PSG), Boro-Silicate Glass (BSG), Boron-Doped Phospho-Silicate Glass (BPSG), or the like. A planarization process such as a CMP process or a mechanical grinding process may be performed to level the top surfaces of ILD **60**, dummy gate stacks **38**, and gate spacers **46** with each other.

FIGS. **7**B and **7**C illustrate the cross-sectional views of an intermediate structure in the formation of a first FinFET and a second FinFET on the same substrate **20** (and in the same die and the same wafer). The cross-sectional views of both of the first FinFET and the second FinFET shown in FIG. **7**B may correspond to the cross-sectional view obtained from the vertical plane containing line A-A in FIG. **7**A. The cross-sectional views of both of the first FinFET and the second FinFET shown in FIG. **7**C may correspond to the cross-sectional view obtained from the vertical plane containing line B-B in FIG. **7**A. In accordance with some embodiments, the first FinFET is a logic device (sometimes referred to as a core device), and is formed in device region **12**-LG. The second FinFET is an input-output (IO) device formed in device region **12**-IO.

After the structure shown in FIGS. **7**A, **7**B, and **7**C is formed, hard mask layers **44** and dummy gate electrodes **42** are removed, forming openings **61** as shown in FIG. **8**A. The respective process is illustrated as process **416** in the process flow **400** as shown in FIG. **24**. The top surfaces and the sidewalls of protruding fins **36** in device regions **12**-LG and **12**-IO are both exposed. Next, an etching mask such as a photo resist **62** is formed in device region **12-10** to protect the dummy gate dielectric **40** in device region **12**-IO. FIG. **8**B illustrates the structure in another cross-section.

In a subsequent process, the dummy gate dielectric **40** in device region **12**-LG is removed, for example, through an isotropic etching process, which may be a dry etching process or a wet etching process. Etching mask **62** (FIGS. **8**A and **8**B) is then removed. The resulting structure is shown in FIGS. **9**A and **9**B. The respective process is illustrated as process **418** in the process flow **400** as shown in FIG. **24**.

FIGS. **10** through **20** illustrate the formation of gate stacks of a FinFET in device region **100** and a FinFET in device region **200**, and the dipole-engineering process in accordance with some embodiments. In accordance with some embodiments, each of device regions **100** and **200** may be selected from a core device region, an IO device region, a memory device region, or the like, in any combination. For example, device region **100** may be a core device region (such as region **12**-LG in FIGS. **9**A and **9**B), while device region **200** may be an IO device region (such as region **12**-IO in FIGS. **9**A and **9**B). Device regions **100** and **200** may also both be core device regions, both be IO regions, both be memory regions, or the like. Furthermore, each of the first FinFET and the second FinFET may be an n-type FinFET or a p-type FinFET in any combination. For example, both of the FinFETs in device regions **100** and **200** may be n-type FinFETs or p-type FinFETs in accordance with some embodiments. In accordance with alternative embodiments of the present disclosure, the FinFET in device region **100** is an n-type FinFET, and the FinFET in device region **200** is a p-type FinFET. Alternatively, the FinFET in device region **100** is a p-type FinFET, and the FinFET in device region **200** is an n-type FinFET. In the subsequent illustrated example, it is assumed that both of device regions **100** and **200** are logic FinFETs, and the corresponding gate dielectrics **40** are replaced with interfacial layers. In accordance with alternative embodiments, one or both of the device regions **100** and **200** are IO device regions. The formation of replacement gate stacks for the IO devices are essentially the same as what are shown in FIGS. **10** through **20**, except gate dielectric **40** is not replaced with interfacial layers.

To distinguish the features in device region **100** from the features in device region **200**, the features in device region **100** may be represented using the reference numerals of the corresponding features in FIG. **7**A plus number **100**, and the features in device region **200** may be represented using the reference numerals of the corresponding features in FIG. **7**A plus number **200**. For example, the source/drain regions **152** and **252** in FIG. **10** correspond to source/drain region **52** in FIG. **7**A, and gate spacers **146** and **246** in FIG. **10** correspond to the gate spacers **46** in FIG. **7**A.

Referring to FIG. **10**, interfacial layers (ILs) **164** and **264** are formed. The respective process is illustrated as process **420** in the process flow **400** as shown in FIG. **24**. ILs **164** and **264** are formed on the top surfaces and the sidewalls of protruding fins **136** and **236**, wherein FIG. **10** illustrates the portions of ILs **164** and **264** on the top surfaces of protruding fins **136** and **236**. In accordance with alternative embodiments, in which one of device region is an IO region, the original gate dielectric **40** (FIG. **9**B) remains, and the subsequently deposited high-k dielectric layer is formed over the original gate dielectric **40**. ILs **164** and **264** may include oxide layers such as silicon oxide layers, which are formed through a thermal oxidation process or a chemical oxidation process to oxidize the surface portions of protruding fins **136** and **236**. ILs **164** and **264** may also be formed through a deposition process. The chemical oxidation process may be performed using a chemical solution (sometimes referred to as Standard Clean 1 (SC1) solution), which comprises NH_4OH, H_2O_2, and H_2O. The chemical oxidation process may also be performed using a Sulfuric Peroxide Mixture (SPM) solution, which is the solution of sulfuric acid and hydrogen peroxide. Alternatively, the chemical oxidation process may be performed using a chemical solution including ozone (O_3) dissolved in water.

In accordance with alternative embodiments, ILs **164** and **264** are formed through thermal oxidation, which may be performed in process gases such as N_2O, O_2, the mixture of N_2O and H_2, the mixture of H_2 and O_2, or the like. The oxidation temperature may be in the range between about 500° C. and about 1,000° C. In accordance with some embodiments, gate dielectric **40** of the IO device has a thickness T**1** (FIG. **9**B) greater than about 15 Å, and may be in the range between about 15 Å and about 50 Å. The thickness T**2** of the replacement ILs (such as the ILs **164** and **264** in FIG. **10**) is smaller than thickness T**1**. In accordance with some embodiments, thickness T**2** is in the range between about 5 Å and about 15 Å.

Next, referring to FIG. **11**, first high-k dielectric layers **166** and **266** are deposited over the corresponding ILs **164** and **264**. The respective process is illustrated as process **422** in the process flow **400** as shown in FIG. **24**. High-k dielectric layers **166** and **266** may be formed of a high-k dielectric material such as hafnium oxide (HfO_2), zirconium oxide (ZrO_2), titanium oxide (TiO_2), or the like, or the combinations thereof such as HfZrO, HfTiO, or the like. The high-k dielectric material may be pure (such as pure HfO_2, pure ZrO_2, or pure TiO_2) or substantially pure (for example, with the atomic percentage being greater than about 90 or 95 percent). The dielectric constant (k-value) of the high-k dielectric material is higher than 3.9, and may be higher than about 7.0. High-k dielectric layers **166** and **266** are overlying, and may physically contact, the respective underlying ILs **164** and **264** (or gate dielectric layers **40**). High-k dielectric layers **166** and **266** are formed as conformal layers, and extend on the sidewalls of protruding fins **136** and **236** and the top surfaces and the sidewalls of gate spacers **146** and **246**, respectively. In accordance with some embodiments of the present disclosure, high-k dielectric layers **166** and **266** are formed using ALD or CVD. The deposition temperature may be in the range between about 200° C. and about 400° C. The thickness T**3** may be in the range between about 6 Å and about 20 Å. First high-k dielectric layers **166** and **266** may be deposited in a common process, and hence are formed of the same material, or may be deposited in different processes, and may be formed of different materials.

Further referring to FIG. **11**, a first dipole film is deposited in a deposition process. The respective process is illustrated as process **424** in the process flow **400** as shown in FIG. **24**. The dipole film includes dipole film (portion) **168** in device region **100**, and dipole film (portion) **268** in device region **200**. Dipole films **168** and **268** are formed through a conformal deposition process such as an ALD process or a CVD process, so that the horizontal thickness of the horizontal portions and the vertical thickness of the vertical portions of dipole films **168** and **268** are substantially equal to each other, for example, with the variation in the thicknesses having a difference smaller than about 20 percent or 10 percent. In accordance with some embodiments of the present disclosure, dipole films **168** and **268** extend into openings **161** and **261**, and include some portions over ILD **60**.

Dipole films **168** and **268** include a dipole-engineering dopant (referred to as dipole dopant hereinafter) such as lanthanum, aluminum, yttrium, Titanium, Magnesium, Niobium, Gallium, Indium or the like. These elements, when diffused into high-k dielectric layers, may increase the number of dipoles, and result in the change in threshold voltages (Vts) of the respective FinFETs. The effect of different dipole dopants on p-type transistors and n-type transistors may be different from each other. For example, La-based dipole dopant will result in the reduction of the Vt of the n-type transistors, and will increase the Vt of p-type transistors. Conversely, Al-based dipole dopant will result in the increase of the Vt of the n-type transistors, and the reduction in the Vt of p-type transistors. Each dipole dopant may exist in both of an n-type transistor and a p-type transistor at the same time, and any combinations of different dipole dopants (as above-mentioned) may exist in an n-type FinFET or a p-type transistor, or in both of a p-type transistor and an n-type transistor at the same time.

The dipole films **168** and **268** may be oxides and/or nitrides of the dipole dopant. For example, the La-containing dipole films **168** and **268** may be in the form of lanthanum oxide (La_2O_3), lanthanum nitride (LaN), or the like, or combinations thereof. The Al-containing dipole films **168** and **268** may be in the form of aluminum oxide (Al_2O_3), aluminum nitride (AlN), or the like, or combinations thereof. The thickness T**4** of dipole films **168** and **268** may be in the range between about 0.3 Å and about 30 Å. It is realized that the thickness T**4** of dipole films **168** and **268** may generally be related to the magnitude of the intended threshold voltage tuning, and the greater threshold voltage tuning is intended, the greater thickness T**4** is.

Referring to FIG. **12**, etching mask **70** is formed and patterned. In accordance with some embodiments, etching mask **70** includes Bottom Anti-Reflective Coating (BARC) **70**A, and photo resist **70**B over BARC **70**A. A hard mask (not shown) may also be added underlying BARC **70**A to assist the etching process. The hard mask may be formed of a metal oxide such as titanium oxide or boron nitride, a metal nitride such as a titanium nitride, or may include a metal nitride layer over a metal oxide layer.

Next, an etching process is performed, in which etching mask **70** is used to remove dipole film **168**. The respective process is illustrated as process **426** in the process flow **400** as shown in FIG. **24**. As a result, high-k dielectric layer **166** is revealed. The resulting structure is shown in FIG. **13**. In accordance with some embodiments of the present disclosure, the etching process is performed through wet etching. For example, when dipole film **168** is formed of the La-based material, an acidic wet etching chemical solution may be adopted. For example, the wet etching chemical may include an acid such as HCl, H_2SO_4, H_2CO_3, HF, for the like, and the acid may be mixed with hydrogen peroxide (H_2O_2) and water, and/or the like. When dipole film **168** is formed of the Al-based material, an alkaline wet etching chemical solution may be adopted. For example, the wet etching chemical may include ammonia (NH_3), hydrogen peroxide (H_2O_2), and water, and/or the like.

Etching mask **70** is then removed, resulting in the structure shown in FIG. **14**, in which dipole film **268** remains over high-k dielectric layer **266**, while no dipole film is over high-k dielectric layer **166**. Further referring to FIG. **14**, drive-in annealing process **72** is performed. The respective process is illustrated as process **428** in the process flow **400** as shown in FIG. **24**. In accordance with some embodiments, annealing process **72** is performed through soak annealing, spike rapid thermal annealing, or the like. When the soak anneal is adopted, the annealing duration may be in the range between about 5 seconds and about 5 minutes. The annealing temperature may be in the range between about 500° C. and about 950° C. The annealing process may be performed in a process gas such as N_2, H_2, NH_3, or the mixture thereof. When the spike rapid thermal annealing process is adopted, the annealing duration may be in the range between about 0.5 seconds and about 3.5 seconds. The annealing temperature may be in the range between about 700° C. and about 950° C. The annealing process may also be performed in a process gas such as N_2, H_2, NH_3, or the mixture thereof. The annealing results in the dipole dopant to be driven into high-k dielectric layer **266**. Throughout the description, the high-k dielectric layer **266** doped with the dipole dopant is referred to as (dipole-dopant containing) high-k dielectric layer **266'**. Due to the nature of diffusion, the highest concentration of the dipole dopant is at the interface between layers **266'** and **268**, and the dopant concentration gradually reduces in the directions of arrows **73**. In accordance with some embodiments, the dosage of the dipole dopant in high-k dielectric layer and the underlying layers is in the range between about 0 atom/cm^2 and about 1E17 atoms/cm^2.

After the drive-in annealing process **72**, dipole film **268** is removed in an etching process. The respective process is illustrated as process **430** in the process flow **400** as shown in FIG. **24**. The etching process may be selected from the same group of candidate processes, and using the same group of candidate etching chemicals, as the etching process shown in FIG. **12**. The details are thus not repeated herein. The resulting structure is shown in FIG. **15**.

In accordance with alternative embodiments and/or in another device region, the process of removing dipole film **168** before the drive-in annealing process **72** is omitted. Accordingly, the dipole dopant in dipole film **168** is also diffused into high-k dielectric **166**. In accordance with these embodiments, both of high-k dielectric layers **166** and **266** are doped with dipole dopants.

FIGS. **16** through **20** illustrate the deposition of a second high-k dielectric layer and a second drive-in annealing process in accordance with some embodiments. It is appreciated that some of the materials and the process details may be the same as the preceding processes shown in FIGS. **11** through **15**. These details are not repeated, and may be found referring to the description of the preceding processes.

Referring to FIG. **16**, high-k dielectric layers **174** and **274** are deposited. The respective process is illustrated as process **432** in the process flow **400** as shown in FIG. **24**. The material of high-k dielectric layers **174** and **274** may be selected from the same group of candidate materials for forming high-k dielectric layers **166** and **266** (FIG. **11**), and may include HfO_2, ZrO_2, TiO_2, or the like, or the combinations thereof such as HfZrO, HfTiO, or the like. High-k dielectric layers **174** and **274** are overlying, and may contact, the respective underlying high-k dielectric layers **166** and **266**. In accordance with some embodiments of the present disclosure, high-k dielectric layers **174** and **274** are formed using ALD or CVD. The deposition temperature may be in the range between about 200° C. and about 400° C. The thickness T**5** may be equal to or smaller than the thickness of the underlying high-k dielectric layers **166** and **266'**. For example, thickness T**5** may be in the range between about 1 Å and about 20 Å.

In accordance with some embodiments, high-k dielectric layers **174** and **274** are formed of a material having a k value lower than the k value of high-k dielectric layer **166**. For example, high-k dielectric layers **174** and **274** may be formed of HfO_2, while high-k dielectric layers **166** and **266** may be formed of ZrO_2 or TiO_2. In accordance with alternative embodiments, high-k dielectric layers **174** and **274** have a same k value, and are formed of a same material, as high-k dielectric layers **166** and **266**. In accordance with yet alternative embodiments, high-k dielectric layers **174** and **274** have a greater k value than high-k dielectric layers **166** and **266**. For example, high-k dielectric layers **174** and **274** may be formed of ZrO_2 or TiO_2, while high-k dielectric layers **166** and **266** may be formed of HfO_2.

Further referring to FIG. **16**, dipole films **176** and **276** are formed through a conformal deposition process such as an ALD process or a CVD process. The respective process is illustrated as process **434** in the process flow **400** as shown in FIG. **24**. Dipole films **176** and **276** include a dipole dopant such as lanthanum (such as La_2O_3 or LaN), aluminum (such as Al_2O_3 or AlN), or the like. The dipole dopant of dipole films **176** and **276** may be the same or different from that of dipole films **168** and **268**. The thickness T**6** of dipole films **176** and **276** may be in the range between about 0.3 Å and about 30 Å.

FIG. **16** further illustrates the formation of etching mask **78**, which may have a structure similar to that of etching mask **70**. The details are thus not repeated herein. In a subsequent process, an etching process is performed to remove dipole film **276**, and hence high-k dielectric layer **274** is exposed, as shown in FIG. **17**. The respective process is illustrated as process **436** in the process flow **400** as shown in FIG. **24**. The etching process may be the same as shown in FIGS. **12** and **13**. Etching mask **78** (shown in FIG. **16**) is then removed, revealing dipole film **176**.

Further referring to FIG. **17**, drive-in annealing process **80** is performed. The respective process is illustrated as process **438** in the process flow **400** as shown in FIG. **24**. The drive-in annealing process **80** is similar to the drive-in annealing process **72** in FIG. **14**, and thus the details are not repeated herein. The dipole dopant in dipole film **176** is diffused into high-k dielectric layer **174**, and possibly high-k dielectric layer **166** with a lower doping concentration than in high-k dielectric layer **174**. In subsequent paragraphs, the high-k dielectric layer **174** incorporating the dipole dopant is referred to as (dipole-dopant containing) high-k dielectric layer **174'**.

After the drive-in annealing process, dipole film **176** is removed in an etching process. The respective process is illustrated as process **440** in the process flow **400** as shown in FIG. **24**. The etching process may be selected from the same group of candidate processes, and using the same group of candidate etching chemicals, as the etching process shown in FIG. **12**. The details are thus not repeated herein. The resulting structure is shown in FIG. **18**.

In accordance with alternative embodiments and/or in another device region, the process of removing dipole film **276** before drive-in annealing process **80** is omitted. Accordingly, the dipole dopant in dipole film **276** is also diffused into high-k dielectric **274**. In accordance with these embodiments, both of high-k dielectric layers **174** and **274** are doped with dipole dopants.

As aforementioned, the k-value of the lower high-k dielectric layers **166/266** may be smaller than, equal to, or greater than, the k-value of the upper high-k dielectric layers **174/274**. Furthermore, dipole dopant doping may be performed on the lower high-k dielectric layer (such as **266**) or upper high-k dielectric layer (such as **174**). Doping lower high-k dielectric layer has different effect in adjusting Vt than doping upper high-k dielectric layer. For example, doping lower high-k dielectric layer may change Vt more than doping upper high-k dielectric layer. In addition, doping a high-k dielectric layer having a lower k-value has different effect in adjusting Vt than doping a high-k dielectric layer having a higher k-value. For example, doping a high-k dielectric layer having a higher k-value may change Vt more than doping a high-k dielectric layer having a lower k-value. Therefore, by selecting whether the upper high-k dielectric layer has a higher, an equal, or a lower k-value (with three possibilities) than the lower high-k dielectric layer, and selecting whether to dope the upper high-k dielectric layer, the lower high-k dielectric layer, or both (with three possibilities), 9 (3×3) potential Vt-adjusting levels are resulted. In accordance with some embodiments, on a same chip, the FinFETs with these different Vt-adjusting levels are formed according to design requirement. In addition, since different dipole dopants such as La and Al also have Vt-adjusting ability different from each other, the Vt-adjusting levels are further multiplied by adopting different dipole dopants for different FinFETs.

FIG. **19** illustrates the formation of gate electrodes **186** and **286**, which includes stacked layers **182** and **282** and possibly metal-filling regions **184** and **284**, respectively. The respective process is illustrated as process **442** in the process flow **400** as shown in FIG. **24**. In accordance with some embodiments of the present disclosure, each of stacked layers **182** and **282** includes an adhesion layer (also known as barrier layer, not shown), which may be formed of TiN, TiSiN, or the like. The stacked layers **182** and **282** also include work function layers, which may include TiN layer, TaN, and/or an Al-based layer (formed of, for example, TiAlN, TiAlC, TaAlN, or TaAlC), depending on whether the respective FinFETs are p-type FinFETs or n-type FinFETs. A blocking layer (not shown) and a filling metal, which are represented by layers **184** and **284**, are then deposited if layers **182** and **282** have not fully fill trenches. Otherwise, layers **184** and **284** are not needed. A planarization process such as a CMP process or a mechanical grinding process is then performed, forming gate electrodes **186** and **286**. Replacement gate stacks **188** and **288**, which include the corresponding gate electrodes **186** and **286** and the corresponding gate dielectrics **164/166/174'** and **264/266'/274** are also formed. FinFETs **190** and **290** are thus formed.

Referring to FIG. **20**, gate stacks **188** and **288** are recessed, and are filled with a dielectric material (such as SiN) to form hard masks **192** and **292**. Etch stop layer **93** is formed over hard masks **192** and **292** and ILD **60**. Etch stop layer **93** is formed of a dielectric material, which may include silicon carbide, silicon nitride, silicon oxynitride, or the like. ILD **94** is formed over etch stop layer **78**, and gate contact plugs **195** and **295** are formed.

FIG. **21** illustrates the distribution of dipole dopants in some portions of the gate stacks shown in FIG. **20**. A magnified view of region **502** in gate stack **188** (FIG. **20**) and a magnified view of region **504** (FIG. **20**) in gate stack **288** are shown in FIG. **21**. The schematic dopant concentrations are shown on the left sides of the corresponding magnified views of regions **502** and **504**. In region **502**, before the formation of stacked metal layers **182**, the peak concentration of dipole concentration occurs at the top surface of high-k dielectric layer **174'**. In subsequent thermal processes, the dipole dopant diffuses upwardly and downwardly, and hence resulting in the dopant profile as shown in FIG. **21**, in which the peak dipole dopant concentration profile **506** is at (or slightly lower than) the top surface of high-k dielectric layer **174'**. The dipole dopant concentration reduces gradually in upward and downward directions. In region **504**, the peak dipole dopant concentration profile **508** is at (or slightly lower than) the top surface of high-k dielectric layer **266'**, and reduces gradually in upward and downward directions.

FIG. **22** illustrates the dopant concentration assuming that when the drive-in annealing process **72** as shown in FIG. **14** is performed, the dipole film **168** (FIG. **12**) is not removed. Accordingly, in region **502**, high-k dielectric layers **166** is also diffused with dipole dopant, and hence forming high-k dielectric layers **166'**. The resulting dipole dopant concentration profiles **510** and **512** are schematically illustrated, with dipole dopant concentration profile **510** representing the dopant of dipole film **168**, which has the peak at (or slightly lower than) the top surface of high-k dielectric layer **166'**. Dipole dopant concentration profile **512** represents the dopant of dipole film **176**, which has the peak at (or slightly lower than) the top surface of high-k dielectric layers **174'**. The total dipole dopant concentration is thus the sum of dipole dopant concentration profiles **510** and **512**. The dipole dopant of profiles **510** and **512** may be the same as each other or different from each other. For example, the dopant of one of profiles **520** and **512** may be La, while the other may be Al. Although La and Al may have opposite effects (with one increasing Vt, and the other reducing Vt), the combination results in an additional Vt level.

FIG. **23** illustrates an example embodiment, in which each of high-k dielectric layers **166** and **174** is formed through a plurality of deposition processes to form a plurality of sub layers. A plurality of dipole-film deposition processes, drive-in annealing processes, and dopant-film removal processes are inserted between the plurality of deposition processes for each sub layer of the high-k dielectric layers **166** and **174**. In accordance with these embodiments, the sub layers of high-k dielectric layer **166** are formed of the same high-k dielectric material and have the same k value. The first dipole dopants of the sub layers of high-k dielectric layer **166** are also the same as each other. Similarly, the sub layers of high-k dielectric layer **174** are formed of the same high-k dielectric material and have the same k value. The second dipole dopants of the sub layers of high-k dielectric layer **174** are also the same as each other. The first dipole dopants may be the same as or different from the second dipole dopants. The profile of the first dipole dopants is shown as **514**, and the profile of the second dipole dopants is shown as **516**. The alternating deposition and drive-in annealing processes may result in a more uniform dipole dopant distribution.

It is appreciated that the aforementioned embodiments, including FIGS. **21**, **22**, and **23**, may co-exist in the same chip and on the same semiconductor substrate **20**. Furthermore, more (such as 1, 2, or 3) high-k dielectric layers may be formed over the dielectric layers shown in FIG. **20**, with each of the high-k dielectric layers doped or not doped by a respective subsequent dipole dopant deposition and drive-in annealing process. This creates more adjustment levels of Vt for different FinFETs on the same chip.

The embodiments of the present disclosure have some advantageous features. By forming multiple high-k dielectric layers having the same k values or different k values, and further by selectively doping dipole dopants for selective ones of the high-k dielectric layers, multiple levels of Vt adjustment may be achieved for different circuit requirements. Through the doping of dipoles, the CET values of the transistors are improved, and the CET scaling feasibility is improved.

In accordance with some embodiments of the present disclosure, a method comprises forming a first oxide layer on a first semiconductor region; depositing a first high-k dielectric layer over the first oxide layer, wherein the first high-k dielectric layer is formed of a first high-k dielectric material; depositing a second high-k dielectric layer over the first high-k dielectric layer, wherein the second high-k dielectric layer is formed of a second high-k dielectric material different from the first high-k dielectric material; depositing a first dipole film over and contacting a first layer selected from the first high-k dielectric layer and the second high-k dielectric layer; performing a first annealing process to drive-in a first dipole dopant in the first dipole film into the first layer; removing the first dipole film; and forming a first gate electrode over the second high-k dielectric layer. In an embodiment, the first dipole film is deposited over and contacting the first high-k dielectric layer. In an embodiment, the first dipole film is deposited over and contacting the second high-k dielectric layer. In an embodiment, the second high-k dielectric layer has a higher k value than the first high-k dielectric layer. In an embodiment, the second high-k dielectric layer has a lower k value than the first high-k dielectric layer. In an embodiment, the method further comprises forming a second oxide layer on a second semiconductor region, wherein both of the first high-k dielectric layer and the second high-k dielectric layer further extend on the second oxide layer; depositing a second dipole film over and contacting a second layer selected from the first high-k dielectric layer and the second high-k dielectric layer, wherein the second layer is different from the first layer, and wherein the second dipole film overlaps the second semiconductor region; performing a second annealing process to drive-in a second dipole dopant in the second dipole film into the second layer; removing the second dipole film; and forming a second gate electrode over the second high-k dielectric layer, wherein the second gate electrode overlaps the second semiconductor region. In an embodiment, the method further comprises, before the second annealing process, removing the second dipole film from a region directly over the first semiconductor region. In an embodiment, the first dipole film comprises a material selected from lanthanum oxide, lanthanum nitride, aluminum oxide, aluminum nitride, or combinations thereof.

In accordance with some embodiments of the present disclosure, a device comprises a first oxide layer on a first semiconductor region; a first high-k dielectric layer comprising a first high-k dielectric material; a second high-k dielectric layer comprising a second high-k dielectric material different from the first high-k dielectric material, wherein the second high-k dielectric layer is overlying and contacts the first high-k dielectric layer; a first dipole dopant in the first high-k dielectric layer and the second high-k dielectric layer, wherein a first peak concentration of the first dipole dopant is at a first top surface of the first high-k dielectric layer or a second top surface of the second high-k dielectric layer; a gate electrode over the second high-k dielectric layer; and a source/drain region on a side of the gate electrode. In an embodiment, the first dipole dopant comprises lanthanum. In an embodiment, the first dipole dopant comprises aluminum. In an embodiment, the first peak concentration of the first dipole dopant is at the first top surface, and the device further comprises a second dipole dopant different from the first dipole dopant, wherein the second dipole dopant has a second peak concentration at the second top surface. In an embodiment, a first one of the first dipole dopant and the second dipole dopant is lanthanum, and a second one of the first dipole dopant and the second dipole dopant is aluminum, and both of lanthanum and aluminum are diffused into each of the first high-k dielectric layer and the second high-k dielectric layer. In an embodiment, the second high-k dielectric layer has a lower k value than the first high-k dielectric layer.

In accordance with some embodiments of the present disclosure, a device comprises a first transistor comprising a first portion of a first high-k dielectric layer; a first portion of a second high-k dielectric layer, wherein the second high-k dielectric layer is over the first high-k dielectric layer, and wherein the first high-k dielectric layer and the second high-k dielectric layer have different k values; a first dipole dopant having a first peak concentration at an interface between the first portion of the first high-k dielectric layer and the first portion of the second high-k dielectric layer; and a second transistor comprising a second portion of the first high-k dielectric layer; a second portion of the second high-k dielectric layer; and a second dipole dopant having a second peak concentration at a top surface of the second high-k dielectric layer. In an embodiment, the first dipole dopant is same as the second dipole dopant. In an embodiment, the first dipole dopant and the second dipole dopant are different from each other. In an embodiment, the first dipole dopant and the second dipole dopant are selected from lanthanum and aluminum. In an embodiment, a first one of the first dipole dopant and the second dipole dopant is lanthanum, and a second one of the first dipole dopant and the second dipole dopant is aluminum. In an embodiment, the first transistor and the second transistor are of a same conductivity type.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A device comprising:

a first semiconductor region;

a first transistor comprising:

a first oxide layer on the first semiconductor region;

a first high-k dielectric layer on the first oxide layer;

a first dipole dopant, wherein a first peak concentration of the first dipole dopant is at a first level of the first high-k dielectric layer;

a second dipole dopant, wherein a second peak concentration of the second dipole dopant is at a second level of the first high-k dielectric layer;

a gate electrode over the first high-k dielectric layer; and

a source/drain region aside of the gate electrode; and

a second transistor comprising:

a second semiconductor region;

a second high-k dielectric layer over the second semiconductor region; and

a third dipole dopant in the second high-k dielectric layer, wherein the first dipole dopant and the third dipole dopant comprise a same element.

2. The device of claim **1**, wherein the first dipole dopant is configured to increase a threshold voltage of the first transistor, and the second dipole dopant is configured to reduce the threshold voltage of the first transistor.

3. The device of claim **1**, wherein both of the first dipole dopant and the second dipole dopant are configured to increase a threshold voltage of the first transistor.

4. The device of claim **1**, wherein both of the first dipole dopant and the second dipole dopant are configured to reduce a threshold voltage of the first transistor.

5. The device of claim **1**, wherein the first dipole dopant comprises lanthanum, and the second dipole dopant comprises aluminum.

6. The device of claim **1**, wherein the first high-k dielectric layer comprises a first high-k dielectric material, and the device further comprises a third high-k dielectric layer comprising a second high-k dielectric material different from the first high-k dielectric material, wherein the third high-k dielectric layer is overlying and contacts the first high-k dielectric layer.

7. The device of claim **6**, wherein the first peak concentration of the first dipole dopant is lower than the third high-k dielectric layer, and the second peak concentration of the second dipole dopant is higher than the first high-k dielectric layer.

8. The device of claim **7**, wherein the second peak concentration of the second dipole dopant is at a top surface of the third high-k dielectric layer.

9. The device of claim **6**, wherein the third high-k dielectric layer has a lower k value than the first high-k dielectric layer.

10. The device of claim **1**, wherein the first transistor is a p-type transistor.

11. The device of claim **1**, wherein the first transistor is an n-type transistor.

12. The device of claim **1**, wherein a third peak concentration of the third dipole dopant is at a top surface of the second high-k dielectric layer.

13. A device comprising:

a transistor comprising:

a first high-k dielectric layer;

a first dipole dopant in the first high-k dielectric layer, wherein the first dipole dopant has a first peak concentration in the first high-k dielectric layer;

a second dipole dopant in the first high-k dielectric layer, wherein the second dipole dopant is different from the first dipole dopant;

a second high-k dielectric layer over the first high-k dielectric layer; and

a gate electrode over the first high-k dielectric layer, wherein the second dipole dopant has a second peak concentration at an interface between the second high-k dielectric layer and the gate electrode.

14. A device comprising:

a semiconductor fin;

a first high-k dielectric layer comprising a first high-k dielectric material, wherein the first high-k dielectric layer is on the semiconductor fin;

a second high-k dielectric layer comprising a second high-k dielectric material different from the first high-k dielectric material, wherein the second high-k dielectric layer is overlying and contacts the first high-k dielectric layer;

a first dipole dopant in the first high-k dielectric layer and the second high-k dielectric layer, wherein the first dipole dopant has a first peak concentration in the first high-k dielectric layer;

a second dipole dopant in the first high-k dielectric layer and the second high-k dielectric layer, wherein the second dipole dopant has a second peak concentration at a top surface of the second high-k dielectric layer;

a gate electrode over the second high-k dielectric layer; and

a source/drain region on a side of the gate electrode.

15. The device of claim **14**, wherein both of the first dipole dopant and the second dipole dopant have a same effect, and the same effect comprises increasing or reducing a threshold voltage of a respective transistor that comprises the gate electrode.

16. The device of claim **14**, wherein both of the first dipole dopant and the second dipole dopant have opposite effects of increasing or reducing a threshold voltage of a respective transistor that comprises the gate electrode.

17. The device of claim **14**, wherein the first dipole dopant and the second dipole dopant comprise lanthanum and aluminum.

18. The device of claim **13**, wherein concentrations of the second dipole dopant gradually reduce in a first direction pointing from the interface toward a top surface of the gate electrode, and gradually reduce in a second direction pointing from the interface toward a bottom surface of the first high-k dielectric layer.

19. The device of claim **13**, wherein the first dipole dopant is configured to increase a threshold voltage of the transistor, and the second dipole dopant is configured to reduce the threshold voltage of the transistor.

20. The device of claim **13**, wherein both of the first dipole dopant and the second dipole dopant are configured to increase a threshold voltage of the transistor.

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